Quantitative Research

Part 3: Statistical Inference

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About the Lecturer



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Professor of Mathematics, specialising in Business Data Science

- Former Technical Lead for Artificial Intelligence and supervisor of industrial doctoral projects at BASF
- Research: Data Science, Machine Learning and Artificial Intelligence
- Focus: Application of AI in business and industry

Course materials:
github.com/ChrisW09/Quantitative-Research-Methods

The lecture at a glance

	Part	Content
	1 – Descriptive Statistics	*Revision:* basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion
	2 – Graphical Representations	*Revision:* empirical CDF, histogram, boxplot
▶	**3 – Statistical Inference**	**Normal distribution, CLT, confidence intervals, sample size, tests, p-values, types of error (appendix: one-sided, power)**
	4 – Regression	Correlation, least squares, goodness of fit, inference for regression, multiple regression (appendix: residual analysis)
	5 – Classification	Conditional probability, logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers
	Python course	Separate course: Python basics, pandas, plots, machine learning (Colab notebooks)

All analyses in **Python** (NumPy, pandas, SciPy, statsmodels, matplotlib); every deck comes with a Python script and a Jupyter notebook – also in Google Colab.

Contents – Part 3

1. Foundations: a Probability Refresher
2. Confidence Intervals
3. Hypothesis Tests
4. The p-Value
5. Types of Error
6. Summary

▶ **Try it:** notebook, all examples and exercises (Colab)

The common thread of this part

One question runs through every section

We draw a **sample** and want to learn something about the **population** from it – while being *honest* about the remaining uncertainty.

One foundation, two tools:

- **Foundations** – "How much does a sample mean fluctuate?" (normal distribution, CLT)
- **Confidence interval** – "Which values of μ are *compatible* with the data?"
- **Hypothesis test** – "Is a specific value μ_0 compatible – or do the data *reject* it?"

One running story:

The dairy

A machine fills milk. Is it working correctly? We measure, estimate and test – and follow this example throughout the entire part.

Learning objectives – Part 3

After this part you will be able to ...

- work with the **normal distribution** and the Φ-table, and explain why $\bar{X}$ is approximately normal (**CLT**),
- compute and correctly interpret a **confidence interval** for μ – and plan the **sample size** it needs,
- carry out a **hypothesis test** (known/unknown variance),
- compute the p-**value** and read it as a measure of evidence,
- distinguish **Type I and Type II errors**,
- read **critical values** from the z- and t-tables and link them to the p-value,
- implement every procedure in Python (`scipy.stats`).

Key takeaway

Common thread: infer from a sample – and report the uncertainty honestly.

→ *Appendix: estimators, law of large numbers, one-sided tests, power*

List of symbols (1/2)

Symbol	Meaning	Symbol	Meaning
$\mu = E(X)$	Expected value	n	Sample size
$\sigma^2 = \mathrm{Var}(X)$	Variance	X_i	Sample variable (random)
σ	Standard deviation	x_i	Observed value
$\bar{X}$	Sample mean (random)	μ_0	Hypothesised value
$\bar{x}$	Realisation of $\bar{X}$	s^2	Sample variance
s	Sample std. dev.	SE	Standard error, $\sigma/\sqrt{n}$
e	Margin of error	H_0	Null hypothesis
H_1	Alternative hypothesis	α	Type I error prob.
β	Type II error prob.	$1-\beta$	Power

Convention: $N(\mu, \sigma^2)$ – the second argument is the **variance** (not the standard deviation).

List of symbols (2/2)

Symbol	Meaning	Symbol	Meaning
$Z \sim N(0,1)$	Standard normal variable	φ, Φ	Density, CDF of $N(0,1)$
Z_n	Standardised mean, σ known (CI pivot / test statistic)	T_n	The same with s instead of σ
z_p	p-quantile: $\Phi(z_p) = p$	$F_{t,n-1}$	CDF of t_{n-1}
$z_{1-\alpha/2}$	Two-sided quantile, $N(0,1)$	$t_{1-\alpha/2,n-1}$	Two-sided quantile, t_{n-1}
$z_{1-\alpha}$	One-sided quantile, $N(0,1)$	$t_{1-\alpha,n-1}$	One-sided quantile, t_{n-1}
p	p-value	CI	Confidence interval
α	Significance level	$1-\alpha$	Confidence level
$\xrightarrow{d}$	Convergence in distribution	i.i.d.	Independent, identically distributed
$\xrightarrow{P}$	Convergence in probability	$\hat{\theta}$	Estimator of a parameter θ

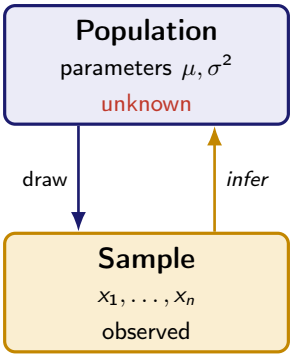
Overview

1. Foundations: a Probability Refresher
   - From data to models
   - Normal distribution
   - Sampling distribution and CLT
2. Confidence Intervals
3. Hypothesis Tests
4. The p-Value
5. Types of Error
6. Summary

Why inferential statistics?

The starting question

We only know one **sample** – but want to say something about the **whole population**.



- **Descriptive:** describes the data at hand.
- **Inferential:** infers *back* from the sample to the population.
- This inference is *uncertain* – we must report the uncertainty.

Two worlds: data and model

Data world (descriptive, Parts 1–2)

A **concrete sample** $x_1, \ldots, x_n$, summarised by

- the mean $\bar{x} = \frac{1}{n}\sum x_i$,
- the variance s^2 and standard deviation s,
- relative frequencies.

Changes from sample to sample.

Model world (probability)

A **random variable** X with an (unknown) distribution:

- the expected value $E(X) = \mu$,
- the variance σ^2, standard deviation σ,
- probabilities.

Fixed – but unknown.

Key takeaway

The sample figures serve as **estimates** of the model parameters: $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = s^2$. Inference asks how far such an estimate can be off.

→ *Appendix: estimators, law of large numbers, one-sided tests, power*

Random variables, expected value and variance

Random variable

A **random variable** X assigns a number to every outcome of a random experiment (e.g. the fill volume of the *next* bottle). Its distribution says which values occur with which probability.

Expected value – probability-weighted mean:

$$\mu = E(X) = \sum_i x_i \, P(X = x_i)$$

Variance – mean squared deviation from μ:

$$\sigma^2 = \mathrm{Var}(X) = E\left[(X-\mu)^2\right] = E(X^2) - \mu^2$$

Continuous X with density f: sums become integrals, e.g. $E(X) = \int x\, f(x)\, \mathrm{d}x$.

Complaints per day in an online shop

x	0	1	2	3
$P(X = x)$	0.1	0.3	0.4	0.2

$E(X) = 0 \cdot 0.1 + 1 \cdot 0.3 + 2 \cdot 0.4 + 3 \cdot 0.2 = \mathbf{1.7}$

$E(X^2) = 0 + 0.3 + 1.6 + 1.8 = 3.7$

$\mathrm{Var}(X) = 3.7 - 1.7^2 = \mathbf{0.81}$, $\sigma = \mathbf{0.9}$

In words: In the long run: 1.7 complaints per day on average – a value that no single day can have.

Two rules that explain why averaging helps

Linear transformation

$$E(a + bX) = a + b\,E(X)$$
$$\mathrm{Var}(a + bX) = b^2\,\mathrm{Var}(X)$$

Sums

$$E(X + Y) = E(X) + E(Y) \quad \text{always}$$
$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \quad \text{if independent}$$

Applied to the sample mean of i.i.d. $X_1, \ldots, X_n$ with $E(X_i) = \mu$, $\mathrm{Var}(X_i) = \sigma^2$:

$$E(\bar{X}) = \frac{1}{n} \cdot n\mu = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}, \qquad \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}.$$

Key takeaway

$\bar{X}$ is right *on average*, and its variance shrinks like $1/n$: the variances of the n values add up to $n\sigma^2$, but averaging divides by n^2. (Variances add – standard deviations do *not*.)

Exercise – expected value and variance: warranty claims

Problem statement

An electronics retailer registers X warranty claims per week. From long experience the distribution is known:

claims per week x	0	1	2	3
$P(X = x)$	0.2	0.4	0.3	0.1

a) Compute $E(X)$, $\mathrm{Var}(X)$ and the standard deviation σ.

b) Each claim costs 150 euros; on top come fixed handling costs of 100 euros per week, i.e. $K = 100 + 150\,X$. Find $E(K)$ and the standard deviation of K.

c) Weeks are independent. For a quarter of $n = 9$ weeks, find the expected value and the standard deviation of the *total* $S = X_1 + \cdots + X_9$ and of the *mean* $\bar{X} = S/9$.

Solution (1/2) – warranty claims: single week and costs

Rules: $E(X) = \sum_i x_i\, P(X = x_i)$; $\mathrm{Var}(X) = E(X^2) - \mu^2$; $E(a + bX) = a + b\,E(X)$, $\mathrm{Var}(a + bX) = b^2\,\mathrm{Var}(X)$.

a) $E(X) = 0 \cdot 0.2 + 1 \cdot 0.4 + 2 \cdot 0.3 + 3 \cdot 0.1 = \mathbf{1.3}$

$E(X^2) = 0 \cdot 0.2 + 1 \cdot 0.4 + 4 \cdot 0.3 + 9 \cdot 0.1 = 2.5$

$\mathrm{Var}(X) = 2.5 - 1.3^2 = 2.5 - 1.69 = \mathbf{0.81}, \qquad \sigma = \sqrt{0.81} = \mathbf{0.9}$

b) $E(K) = 100 + 150 \cdot 1.3 = 100 + 195 = \mathbf{295\ \text{euros}}$

$\mathrm{Var}(K) = 150^2 \cdot 0.81 = 18\,225, \qquad \sigma_K = 150 \cdot 0.9 = \mathbf{135\ \text{euros}}$

In words: On average 1.3 claims and 295 euros of warranty costs per week, typically fluctuating by about ± 135 euros. The fixed 100 euros shift the costs but do not spread them.

Solution (2/2) – warranty claims: total and mean of 9 weeks

Rules: expected values always add; for *independent* weeks the *variances* add; $\bar{X} = S/n \Rightarrow \mathrm{Var}(\bar{X}) = \sigma^2/n$.

c) $E(S) = 9 \cdot 1.3 = \mathbf{11.7}$, $\mathrm{Var}(S) = 9 \cdot 0.81 = 7.29$, $\mathrm{sd}(S) = \sqrt{7.29} = \mathbf{2.7}$

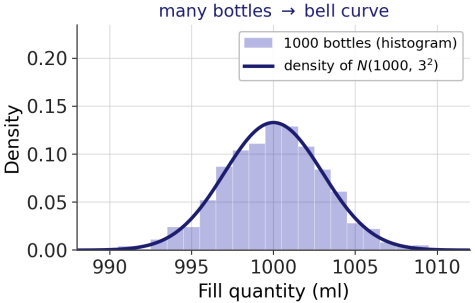
$E(\bar{X}) = \mathbf{1.3}$, $\mathrm{Var}(\bar{X}) = 0.81/9 = 0.09$, $\mathrm{sd}(\bar{X}) = 0.9/\sqrt{9} = \mathbf{0.3}$

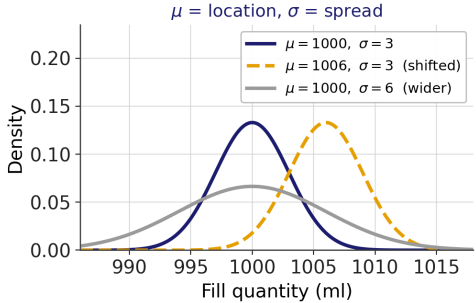
In words: The total spreads $\sqrt{9} = 3$ times as much as one week – not 9 times ($9 \cdot 0.9 = 8.1$ is the typical mistake: variances add, standard deviations do not). The mean spreads only a third as much; this $\sigma/\sqrt{n}$ returns below as the **standard error**.

```
import numpy as np
x = np.array([0, 1, 2, 3]); p = np.array([0.2, 0.4, 0.3, 0.1])
EX = np.sum(x * p); VarX = np.sum(x ** 2 * p) - EX ** 2
print(round(EX, 2), round(VarX, 2))                                  # a)  -> 1.3 0.81
print(round(np.sqrt(9 * VarX), 2), round(np.sqrt(VarX / 9), 2))      # c)  -> 2.7 0.3
```

▶ **Try it:** notebook, section 1.1 (Colab)

The normal distribution $N(\mu, \sigma^2)$





Density of $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$$

- $E(X) = \mu$ (location), $\mathrm{Var}(X) = \sigma^2$ (spread); symmetric bell around μ.
- Probabilities are **areas** under f.
- Careful: `scipy` wants the *standard deviation*, `stats.norm(loc=`μ`, scale=`σ`)`.

Standardisation: one table for all normal distributions

Standard normal distribution

$Z \sim N(0,1)$ has density $\varphi(z)$ and CDF $\Phi(z) = P(Z \le z)$. Normal areas have no closed formula, but every normal variable can be **standardised**:

$$X \sim N(\mu, \sigma^2) \quad \Longrightarrow \quad Z = \frac{X - \mu}{\sigma} \sim N(0,1), \qquad \text{conversely } X = \mu + \sigma Z.$$

Milk: a single bottle, $X \sim N(1000,\, 3^2)$

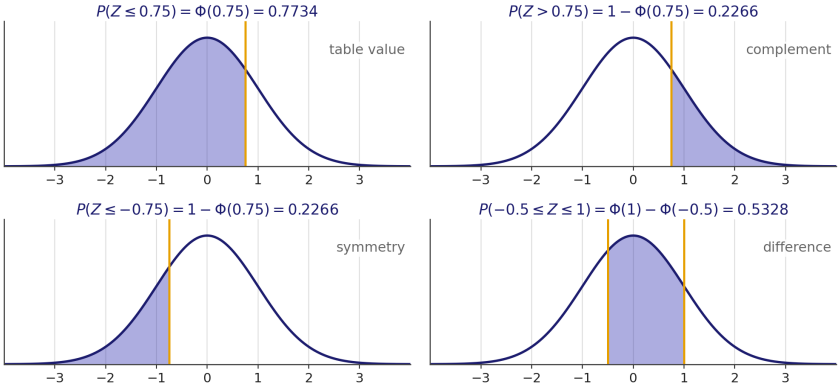
$$P(X \le 997) = P\left(Z \le \tfrac{997-1000}{3}\right) = \Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = \mathbf{0.1587}$$

997 ml lies one standard deviation below μ: about 16 % of the bottles hold at most 997 ml.

Key takeaway

Standardising reduces every normal probability to *one* table – that of Φ.

Probabilities as areas – the four basic cases



The table gives only $\Phi(z) = P(Z \le z)$. Everything else follows from **total area** $= 1$ and **symmetry**: $P(Z > z) = 1 - \Phi(z)$, $\Phi(-z) = 1 - \Phi(z)$, $P(a \le Z \le b) = \Phi(b) - \Phi(a)$.

Standard normal table $\Phi(z) = P(Z \leq z)$ (1/2)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

Reading: row $=$ z to one decimal, column $=$ second decimal. Highlighted: $z = 0.75 \rightarrow$ row 0.7, column .05 $\rightarrow$ $\Phi(0.75) = 0.7734$.

Standard normal table $\Phi(z) = P(Z \leq z)$ (2/2)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

Negative z via symmetry: $\Phi(-z) = 1 - \Phi(z)$. For $z \geq 3$: $\Phi(z) \approx 1$. Highlighted: the quantiles 1.645 and 1.96 (next slide).

Reading the table – forwards and backwards

Forwards: probability wanted

$z = 0.75$: row 0.7, column .05 $\Rightarrow$
$\Phi(0.75) = \mathbf{0.7734}$.
Negative z: $\Phi(-0.75) = 1 - 0.7734 = 0.2266$.

Backwards: quantile wanted

Search 0.95 *inside* the table: $\Phi(1.64) = 0.9495$,
$\Phi(1.65) = 0.9505 \Rightarrow z_{0.95} \approx \mathbf{1.645}$.
$\Phi(1.96) = 0.9750 \Rightarrow z_{0.975} = \mathbf{1.96}$.

The p-**quantile** z_p is defined by $\Phi(z_p) = p$; by symmetry $z_{1-p} = -z_p$ (e.g. $z_{0.05} = -1.645$).

```
from scipy import stats
stats.norm.cdf(0.75)                  # Phi(0.75)                -> 0.7734
stats.norm.ppf(0.95)                  # quantile z_0.95          -> 1.6449
stats.norm.cdf(997, loc=1000, scale=3)   # P(X <= 997), N(1000,9) -> 0.1587
```

Key takeaway

The famous 1.96 is the 0.975-quantile of $N(0,1)$: the central 95 % of the area lies between ± 1.96.

▶ **Try it:** notebook, section 1.2 (Colab)

Exercise – normal distribution: fill volume

Problem statement

A bottling plant fills orange juice into 1-litre bottles. The fill volume X (in ml) is normally distributed with $\mu = 1000$ and $\sigma = 10$, i.e. $X \sim N(1000, 10^2)$. The legal minimum content is 985 ml.

a) What share of the bottles is underfilled, $P(X < 985)$?

b) With what probability does a bottle contain between 990 and 1015 ml?

c) Which fill volume $x_{0.95}$ is exceeded by only 5 % of the bottles?

d) The share of underfilled bottles is to be cut to at most 1 %. To which setpoint μ must the plant be adjusted if $\sigma = 10$ stays the same?

Hint: standardise, $z = (x - \mu)/\sigma$, and use the Φ table.

Solution (1/3) – fill volume: probabilities

Rules: $P(X \le x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$; $\Phi(-z) = 1 - \Phi(z)$; $P(a \le X \le b) = \Phi(z_b) - \Phi(z_a)$.

a) $P(X < 985) = \Phi\left(\frac{985-1000}{10}\right) = \Phi(-1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = \mathbf{0.0668}$

b) $P(990 \le X \le 1015) = \Phi\left(\frac{1015-1000}{10}\right) - \Phi\left(\frac{990-1000}{10}\right) = \Phi(1.5) - \Phi(-1)$

$= 0.9332 - (1 - 0.8413) = 0.9332 - 0.1587 = \mathbf{0.7745}$

Interpretation

About 6.7 % of the bottles are underfilled; about 77.5 % contain between 990 and 1015 ml.

Solution (2/3) – fill volume: quantile and setpoint

Rules: quantile $x_p = \mu + \sigma z_p$ (search p *inside* the table); $z_{1-p} = -z_p$.

c) $\Phi(z) = 0.95 \Rightarrow z_{0.95} = 1.645 \Rightarrow x_{0.95} = 1000 + 1.645 \cdot 10 = \mathbf{1016.45\ ml}$

d) $P(X < 985) = 0.01 \Rightarrow \frac{985-\mu}{10} = z_{0.01} = -z_{0.99} \approx -2.33$ (table: $\Phi(2.33) = 0.9901$)
$\Rightarrow \mu = 985 + 2.33 \cdot 10 = \mathbf{1008.3\ ml}$ (exact $z_{0.99} = 2.3263$: 1008.26 ml)

Interpretation

Only 5 % of the bottles contain more than 1016.45 ml. To cut underfilling from 6.7 % to 1 %, the setpoint must rise by about 8 ml – every bottle gets more juice: the price of safety.

Solution (3/3) – fill volume: with Python

Rule: `cdf` gives $P(X \le x)$, `ppf` the quantile; `scale` $= \sigma$ is the *standard deviation*.

```
from scipy import stats

X = stats.norm(loc=1000, scale=10)          # X ~ N(1000, 10^2)
print(f"a) {X.cdf(985):.4f}")
print(f"b) {X.cdf(1015) - X.cdf(990):.4f}")
print(f"c) {X.ppf(0.95):.2f}")
print(f"d) {985 - stats.norm.ppf(0.01) * 10:.2f}")    # mu = 985 - z_0.01 * sigma
```

Output

```
a) 0.0668
b) 0.7745
c) 1016.45
d) 1008.26
```

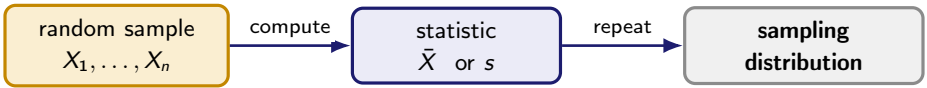
The table answers differ only by rounding ($z_{0.99} = 2.33$ instead of 2.3263 in d).

▶ **Try it:** notebook, section 1.2 (Colab)

Estimators are random variables

The sample $X_1, \ldots, X_n$ is **random**, so *anything* computed from it – $\bar{X}$, s, and later the test statistics – is itself a **random variable**.

- New sample $\Rightarrow$ a new $\bar{x}$ *and* a new s.
- Its distribution over *all possible samples* is the **sampling distribution** – so $\bar{X}$ and s each have their own.



Key takeaway

A confidence interval or test is built on the *sampling distribution* of an estimator – never on a single number alone.

The standard error: how much $\bar{X}$ wobbles

Definition

The **standard error** (SE) is the *standard deviation of the sampling distribution* of an estimator – the spread of $\bar{X}$ from sample to sample, *not* the spread of the raw data.

$$\mathrm{SE}(\bar{X}) = \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}} \quad \left(\text{or } \frac{s}{\sqrt{n}} \text{ when } \sigma \text{ is estimated}\right).$$

- It is the natural **ruler** for inference: we will measure *how many standard errors* $\bar{x}$ lies away from a reference value – this standardised quantity reappears as the test statistic in Section 3.
- Larger $n \Rightarrow$ smaller SE $\Rightarrow$ narrower CI and sharper test.

Key takeaway

Standard *deviation* = spread of the data; standard *error* = spread of the estimate.

The central limit theorem – intuition

Guiding question

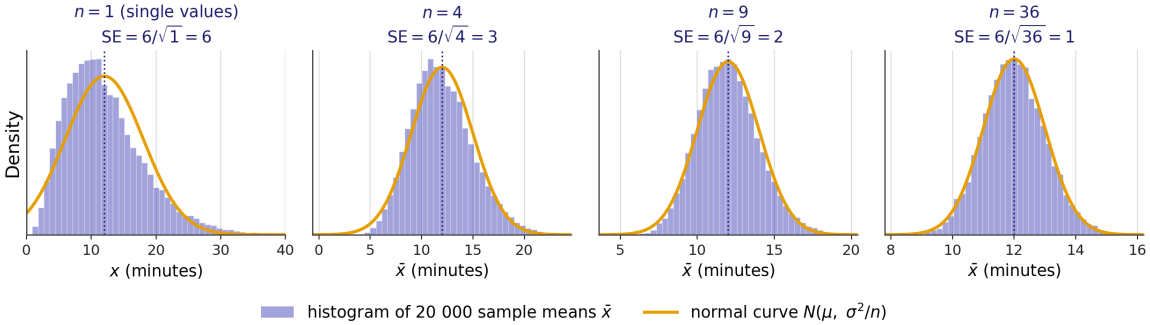
Our inference rests on $\bar{X} \sim N(\mu, \sigma^2/n)$. But real data – waiting times, incomes, processing times – are often *skewed*. Does the recipe still work?

- Draw n independent values from *any* distribution (skewed, flat, bimodal) with mean μ and standard deviation σ.
- Their **mean** $\bar{X}$ is approximately **normal** once n is large enough – the shape of the original distribution washes out.
- Why? In an average, unusually high and unusually low values partly cancel out; what remains is a symmetric bell around μ.

Key takeaway

Whatever the shape of the data: for large n the **sample mean** is approximately normal, with centre μ and standard error $\sigma/\sqrt{n}$.

The central limit theorem – a simulation



Processing times of customer orders are right-skewed (here: gamma distribution with $\mu = 12$, $\sigma = 6$ minutes). For each n, 20 000 samples were drawn and their means $\bar{x}$ plotted. For $n = 1$ the normal curve fits badly; at $n = 36$ the histogram of $\bar{x}$ is practically normal – and it narrows like $6/\sqrt{n}$.

The central limit theorem – formal statement

Central limit theorem (Lindeberg–Lévy)

Let $X_1, \ldots, X_n$ be i.i.d. with $E(X_i) = \mu$ and $0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$. Then

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \quad (n \to \infty), \qquad \text{i.e.} \qquad \bar{X} \overset{\text{approx.}}{\sim} N\Big(\mu, \frac{\sigma^2}{n}\Big).$$

In words: For large n the standardised mean behaves like a standard normal variable – whatever the distribution of the single values.

- $\xrightarrow{d}$ = convergence *in distribution*: the CDF of Z_n approaches Φ.
- Rule of thumb: $n \gtrsim 30$ is usually enough; strongly skewed data need more.
- For *normal* data, $\bar{X} \sim N(\mu, \sigma^2/n)$ holds *exactly*, for every n.
- The CLT is about $\bar{X}$ – it does *not* make the single values normal.

The central limit theorem with Python

Simulation: 20 000 shifts with $n = 36$ right-skewed processing times each (gamma distribution with $\mu = 12$, $\sigma = 6$ minutes, as on the simulation slide).

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(42)                                  # fixed seed
times = rng.gamma(shape=4, scale=3, size=(20_000, 36))    # 20 000 shifts x 36 orders
means = times.mean(axis=1)                                       # one mean per shift
times.mean(), times.std()                          # single values:  12.0, 6.0
means.mean(), means.std()                          # shift means:    12.003, 0.99
stats.skew(times.ravel()), stats.skew(means)       # skewness:       1.0, 0.15
np.mean(means > 13.5)                              # share of shifts above 13.5 min: 0.0698
stats.norm.sf(13.5, loc=12, scale=6 / np.sqrt(36))       # CLT approximation:  0.0668
```

In words: The shift means are centred at $\mu = 12$, spread with $\sigma/\sqrt{n} = 1$ and are almost symmetric (skewness $1.0 \to 0.15$), although every single time is right-skewed. The simulated share above 13.5 minutes is close to the CLT value.

▶ **Try it:** notebook, section 1.3 (Colab)

Exercise – central limit theorem: processing times

Problem statement

The processing time X of a customer order in a service centre (in minutes) is **right-skewed**; its exact distribution is unknown. Known are only $\mu = 12$ and $\sigma = 6$. In one shift, $n = 36$ orders are processed independently of each other.

a) How is the mean processing time $\bar{X}$ of a shift (approximately) distributed? Why?

b) With what probability is the mean processing time of a shift above 13.5 minutes?

c) What would $P(X > 13.5)$ be for a *single* order if X were normal? Why can this not be computed without that assumption – whereas b) can?

d) How many orders per shift would be needed for a standard error of the mean of at most 0.5 minutes?

Solution (1/3) – processing times: distribution of the mean

Rule (CLT): for i.i.d. X_i and large n, $\bar{X} \overset{\text{approx.}}{\sim} N(\mu, \sigma^2/n)$ – whatever the shape of the distribution of X.

a) $E(\bar{X}) = 12, \qquad \text{SE} = \dfrac{\sigma}{\sqrt{n}} = \dfrac{6}{\sqrt{36}} = 1 \quad \Rightarrow \quad \bar{X} \overset{\text{approx.}}{\sim} \mathbf{N(12, 1^2)}$

b) $P(\bar{X} > 13.5) = P\left(Z > \frac{13.5-12}{1}\right) = 1 - \Phi(1.5) = 1 - 0.9332 = \mathbf{0.0668}$

Interpretation

Although single processing times are skewed, the mean of 36 orders is approximately normal ($n = 36 \geq 30$). Only in about 6.7 % of the shifts does the mean processing time exceed 13.5 minutes.

Solution (2/3) – processing times: single value vs. mean

Rule: $P(X > x) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right)$ for a *single* value holds **only** if X itself is normal – the CLT says nothing about X, only about $\bar{X}$.

c) $$P(X > 13.5) = 1 - \Phi\left(\frac{13.5-12}{6}\right) = 1 - \Phi(0.25) = 1 - 0.5987 = \mathbf{0.4013}$$

(valid *only* if X itself is normal)

- For a skewed X with the same μ and σ the true value can differ markedly. Example: under the gamma model of the simulation slide, $P(X > 13.5) = 0.342$ – not 0.401.
- $P(\bar{X} > 13.5) = 0.0668$ is reliable: the CLT delivers the normal shape of the mean *without* knowing the shape of X (gamma model, exact: 0.071).

Key takeaway

A single order exceeds 13.5 minutes often (roughly 35–40 %), the shift mean rarely (≈ 7 %): averaging shrinks the spread by the factor $\sqrt{36} = 6$.

Solution (3/3) – processing times: sample size and Python

Rule: $\mathrm{SE} = \sigma/\sqrt{n}$ – the standard error falls with $\sqrt{n}$, not with n.

d) $$\frac{6}{\sqrt{n}} \le 0.5 \;\Leftrightarrow\; \sqrt{n} \ge \frac{6}{0.5} = 12 \;\Leftrightarrow\; n \ge \mathbf{144}$$

In words: Halving the standard error (from 1 to 0.5 minutes) requires four times as many orders (36 → 144).

Output (code below)

```
b) 0.0668
c) 0.4013
d) 144
```

```
import numpy as np
from scipy import stats

print(f"b) {stats.norm.sf(13.5, loc=12, scale=6 / np.sqrt(36)):.4f}")  # mean
print(f"c) {stats.norm.sf(13.5, loc=12, scale=6):.4f}")    # single value, IF normal
print(f"d) {(6 / 0.5) ** 2:.0f}")
```

▶ **Try it:** notebook, section 1.3 (Colab)

Overview

1 Foundations: a Probability Refresher

2 Confidence Intervals
- Motivation and intuition
- Known variance
- Length of the confidence interval
- Unknown variance

3 Hypothesis Tests

4 The p-Value

5 Types of Error

6 Summary

A single estimate is not enough

Intuition

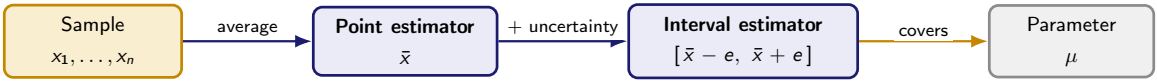
From the sample we compute the mean $\bar{x}$. But it almost **never** hits the true value μ exactly – a second sample would give a different value.

- Instead of a single point we report a **range** of values that are *compatible with the data*.
- The more precise our estimate, the *narrower* this range.

Key takeaway

A **confidence interval** reports the estimate *together with its uncertainty* – a range of plausible values for μ. (It is *not* a probability statement about where μ lies – see the interpretation slides.)

From the sample to the interval



- The point estimator $\bar{x}$ is the *best single estimate*.
- The *margin of error* $e = z_{1-\alpha/2}\,\sigma/\sqrt{n}$ (half the interval's width) turns it into an interval – built so that the *procedure* covers the fixed true μ in $(1-\alpha)$ of samples.

Why divide by the standard error?

The raw deviation is not enough

A sample mean $\bar{x} = 8$ lies 2 units from a reference value $\mu_0 = 10$. Whether that gap is *small or large* depends entirely on the standard error – how much $\bar{X}$ wobbles.

Scenario	$\text{SE} = \sigma/\sqrt{n}$	$z = \dfrac{8-10}{\text{SE}}$	Reading
large spread, small n $(\sigma = 6,\ n = 9)$	2.0	-1.0	ordinary
small spread, large n $(\sigma = 4,\ n = 100)$	0.4	-5.0	far out

Key takeaway

Same raw deviation, opposite readings. Dividing by the SE turns it into a *universal* ruler – "how many standard errors" – comparable across samples and units.

Exercise – standard error: packing flour

Problem statement

A mill packs flour into 500 g bags. The single bags scatter with a known standard deviation of $\sigma = 8\,\text{g}$.

a) What is the standard error of $\bar{X}$ for $n = 16$ and for $n = 64$ bags?

b) A sample gives $\bar{x} = 497\,\text{g}$. How many standard errors does $\bar{x}$ lie below the target $\mu_0 = 500\,\text{g}$ – for $n = 16$ and for $n = 64$? Which finding is conspicuous?

c) How much do the *single* bags scatter if 64 instead of 16 bags are drawn?

Hint: the central 95 % of $N(0,1)$ lie between ± 1.96; $\Phi(1.50) = 0.9332$.

Solution – standard error: packing flour

Rule: $\text{SE} = \sigma/\sqrt{n}$; distance in standard errors: $z = (\bar{x} - \mu_0)/\text{SE}$.

	$n = 16$	$n = 64$
a) $\text{SE} = \sigma/\sqrt{n}$	$8/\sqrt{16} = 8/4 = \mathbf{2\ g}$	$8/\sqrt{64} = 8/8 = \mathbf{1\ g}$
b) $z = (497 - 500)/\text{SE}$	$-3/2 = \mathbf{-1.5}$	$-3/1 = \mathbf{-3.0}$
reading	$\lvert z\rvert < 1.96$: ordinary fluctuation	$\lvert z\rvert > 1.96$: conspicuous

For $n = 16$, a deviation of at least 1.5 standard errors occurs in $2\,(1 - \Phi(1.50)) = 2 \cdot 0.0668 = 0.1336$, i.e. in about 13 % of all samples; 3 standard errors lie beyond the central 95 %.

c) Unchanged, $\sigma = \mathbf{8\ g}$ – a larger n shrinks only the spread of the *mean*, not that of the data.

Key takeaway

Same 3 g shortfall, opposite verdicts: it is the distance in **standard errors** that counts.

How we read probability: the frequentist view

Fixed vs. random

- Parameters μ, σ are **fixed** (unknown) constants – not random.
- The data – and all we build from them ($\bar{X}$, s, Z_n, the CI bounds, later the test results) – are **random**.

What "probability" means here

The long-run *relative frequency* of an event over (hypothetically) very many repeated samples.

Key takeaway

"95 % confidence" and "5 % significance" describe the **procedure across repeated samples** – not one particular interval and not one particular dataset.

→ *Appendix: law of large numbers – why relative frequencies settle down*

Our story: filling milk

Example with known variance

- A dairy fills milk (label: $1\,\text{l} = 1000\,\text{ml}$).
- Machine with **known** variance $\sigma^2 = 9\,\text{ml}^2$, i.e. $\sigma = 3\,\text{ml}$.
- After servicing: is it still working correctly?
- Sample $n = 9$ (in ml):

1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009

Key figures

$$\bar{x} = 1001\,\text{ml}$$

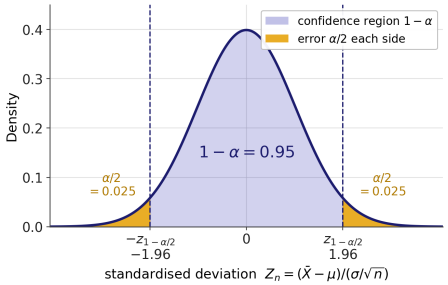
$$1 - \alpha = 0.95 \;\Rightarrow\; \alpha = 0.05$$

Goal: confidence interval for the true mean μ.

How does the mean fluctuate? – graphically

Recall from Foundations

Over repeated samples, $\bar{x}$ scatters around μ with standard error $\sigma/\sqrt{n}$ (milk: $3/\sqrt{9} = 1$ ml), normally – exactly for normal data, approximately by the CLT. Standardised: $Z_n \sim N(0,1)$.



The central (blue) region holds 95 % of the probability – in 95 % of samples Z_n lands here; 2.5 % in each tail.

How does the mean fluctuate? – formally

Sampling distribution of the sample mean

For $X_i \sim N(\mu, \sigma^2)$ i.i.d., the *sampling distribution* of the random variable $\bar{X}$ is

$$\bar{X} \sim N\Big(\mu,\ \tfrac{\sigma^2}{n}\Big) \quad \Longrightarrow \quad Z_n := \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Both results are from the Foundations ($\mathrm{Var}(\bar{X}) = \sigma^2/n$, standardisation). New is the probability statement below – the building block of the confidence interval.

Two-sided probability statement

$$P\big(-z_{1-\alpha/2} \le Z_n \le z_{1-\alpha/2}\big) = 1 - \alpha$$

In words: With probability $1 - \alpha$, Z_n lies between the two quantiles $\pm z_{1-\alpha/2}$.

Derivation (1/2): rearrange step by step

Start from the pivot $Z_n = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$, which lies within $\pm z_{1-\alpha/2}$ with probability $1-\alpha$:

$$1-\alpha = P\Big(-z_{1-\alpha/2} \;\le\; \tfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \;\le\; z_{1-\alpha/2}\Big) \qquad \triangleright \text{ starting point}$$

$$= P\Big(-z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\le\; \bar{X} - \mu \;\le\; z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ multiply all 3 parts by } \tfrac{\sigma}{\sqrt{n}} > 0$$

$$= P\Big(-\bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\le\; -\mu \;\le\; -\bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ subtract } \bar{X} \text{ everywhere}$$

In words: Multiplying by a *positive* number keeps the "$\le$" signs; we only used equivalence steps, so the probability stays exactly $1-\alpha$.

Derivation (2/2): from $-\mu$ to the interval

$$1-\alpha = P\Big(\bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\geq\; \mu \;\geq\; \bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ multiply by } (-1): \text{ signs flip}$$

$$= P\Big(\bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\leq\; \mu \;\leq\; \bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ just read it left-to-right}$$

Key takeaway

Confidence interval with **known** variance:

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

Multiplying an inequality by -1 **reverses** "$\leq$" into "$\geq$" – that is the only subtle step. What is random are the *bounds* (they contain $\bar{X}$); μ itself is a fixed unknown.

The formula in words

$$\text{CI} = \Big[\underbrace{\bar{X}}_{\text{estimate}} \pm \underbrace{z_{1-\alpha/2}}_{\substack{\text{critical} \\ \text{value}}} \cdot \underbrace{\frac{\sigma}{\sqrt{n}}}_{\text{standard error}} \Big]$$

- **Centre** $\bar{X}$: our best estimate of μ.
- **Standard error** $\sigma/\sqrt{n}$: how strongly $\bar{X}$ fluctuates.
- **Critical value** $z_{1-\alpha/2}$: how many standard errors we go to the left/right (1.96 at 95 %).

Milk example: calculation

Given: $\bar{x} = 1001$, $\sigma = 3$, $n = 9$, $\alpha = 0.05$, $z_{0.975} = 1.96$.

$$\text{margin of error } e = z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{3}{\sqrt{9}} = 1.96$$

$$\text{lower bound} = 1001 - 1.96 = 999.04$$

$$\text{upper bound} = 1001 + 1.96 = 1002.96$$

Result

$$\text{CI}_{0.95} = \big[\, 999.04\,;\, 1002.96 \,\big] \text{ ml}$$

The hypothesised target value $\mu_0 = 1000$ lies *inside* the interval $\Rightarrow$ no evidence against a correctly working machine.

Exercise – filling cream (known variance)

Problem statement

A dairy fills cream into cups (label: 250 g). The variance of the machine is known: $\sigma^2 = 4\,\text{g}^2$. After servicing, $n = 16$ cups are sampled (in g):

$$251, 248, 248, 249, 250, 251, 247, 246, 252, 251, 248, 247, 246, 249, 248, 252$$

Confidence level $1 - \alpha = 0.95$.

Required: the confidence interval for μ with *known* variance. Does the target value 250 g lie inside – and what does that say about the machine?

Hint: $z_{0.975} = 1.96$. (The same cups return with *unknown* variance at the end of this section.)

Solution – filling cream (known variance)

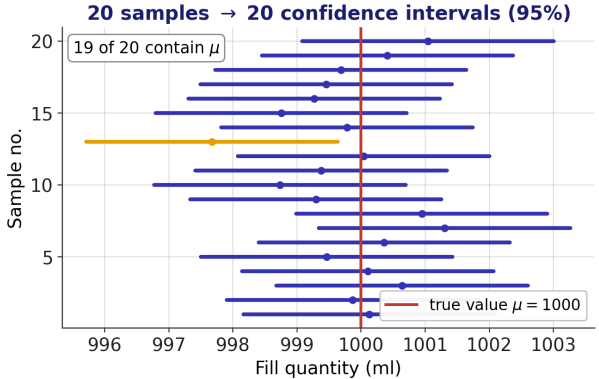
Rule: $\mathrm{CI} = \left[\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}\right]$.
Intermediate values: $\bar{x} = 3983/16 = 248.9375$, $\sigma = 2$, $n = 16$, $z_{0.975} = 1.96$.

$$e = 1.96 \cdot \frac{2}{\sqrt{16}} = 1.96 \cdot 0.5 = 0.98$$
$$\mathrm{CI}_{0.95} = [248.9375 \pm 0.98] = [\mathbf{247.96}\,;\,\mathbf{249.92}]$$

Interpretation

The target value $\mu_0 = 250$ lies *outside* the CI $\Rightarrow$ with known variance: indication that the machine is **not** working correctly.

What does "95 % confidence" mean?



Of 20 computed 95 % intervals, on average ≈ 19 cover the true value μ (red line). The confidence *level* refers to the **procedure** across many samples.

A common misconception

Careful with the interpretation

"With 95 % probability μ lies in *this particular* interval" is, strictly speaking, **wrong**.

- μ is a fixed (unknown) number – it either is or is not inside the particular interval.
- What is random are the *bounds* of the interval, not μ.
- Correct: "The **procedure** yields, in 95 % of samples, an interval that covers μ."

Three correct ways to read a confidence interval

1. **Coverage (the procedure).** The method produces intervals that contain the fixed true μ in a fraction $(1 - \alpha)$ of all samples.
2. **Compatibility.** The interval is the range of values for μ that are *consistent with the observed data*.
3. **Duality with the test.** It is exactly the set of null values μ_0 that a two-sided level-α test would *not* reject.

Key takeaway

All three avoid the trap of putting a probability on *this one* interval: μ is fixed, the **interval** is what is random.

Exercise – reading a confidence interval correctly

Problem statement

For the cream sample ($n = 16$, $\bar{x} = 248.9375\,\text{g}$, $\sigma = 2\,\text{g}$ known) the 95 % CI $[247.96\,;\,249.92]$ was computed. Which of the following statements are **true**, which are **false**? Give a short reason.

1. "With 95 % probability the true mean μ lies between 247.96 g and 249.92 g."
2. "About 95 % of all filled cups weigh between 247.96 g and 249.92 g."
3. "If the sample were repeated very often, about 95 % of the intervals computed this way would contain the true value μ."
4. "The target value 250 g is *not* compatible with the data at the 95 % level."
5. "A 99 % interval from the same data would be *narrower*, because it is more certain."

Solution (1/2) – reading a confidence interval correctly

Rule: μ is a *fixed* number; the *bounds* are random. The confidence level describes the *procedure*, not the one interval.

Answers 1–3

1. **False.** μ either lies in this particular interval or it does not – the probability 0.95 belongs to the procedure across many samples.
2. **False.** The CI describes the uncertainty about μ (spread $\sigma/\sqrt{n} = 0.5$), not the spread of single cups ($\sigma = 2$). The fluctuation interval of the cups would be $248.9375 \pm 1.96 \cdot 2 = [\mathbf{245.02}; \mathbf{252.86}]$ – four times as wide.
3. **True.** That is exactly the coverage reading: the procedure hits μ in 95 % of the samples.

Solution (2/2) – reading a confidence interval correctly

Answers 4–5

4. **True.** $250 \notin [247.96\,;\,249.92]$ – the target is not among the values compatible with the data ("compatibility" reading). A two-sided test at level 5 % would reject $H_0 : \mu = 250$ (next section).
5. **False.** More confidence costs precision: $z_{0.995} = 2.576 > 1.96$, so $e = 2.576 \cdot \frac{2}{\sqrt{16}} = 1.288$ and $\text{CI}_{0.99} = [248.9375 \pm 1.288] = [\mathbf{247.65}\,;\,\mathbf{250.23}]$ – *wider* than the 95 % CI, and now 250 even lies *inside*.

Key takeaway

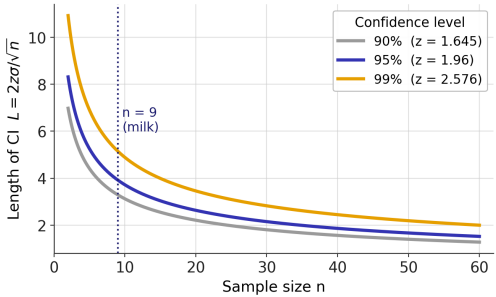
A CI is a statement about the **procedure** and about the values of μ **compatible** with the data – not about single values and not about "the probability of μ".

What does the length depend on?

Length

$$L = 2 \cdot z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- Higher confidence level $\Rightarrow$ larger $z_{1-\alpha/2}$ $\Rightarrow$ *longer*.
- Larger $n \Rightarrow$ *shorter* (with $1/\sqrt{n}$).
- Larger spread $\sigma \Rightarrow$ *longer*.



In words: More confidence *or* more precision – both require a larger sample.

Different confidence levels – milk

Confidence level	Quantile $z_{1-\alpha/2}$	CI for milk example
90 %	$z_{0.95} = 1.645$	[999.36 ; 1002.64]
95 %	$z_{0.975} = 1.96$	[999.04 ; 1002.96]
99 %	$z_{0.995} = 2.576$	[998.42 ; 1003.58]

Key takeaway

Higher confidence level $\Rightarrow$ wider interval $\Rightarrow$ less precise statement. Confidence and precision are a trade-off.

Planning the sample size

Guiding question

How many bottles must the dairy measure so that the 95 % CI is at most $\pm e$ wide – e.g. ± 1 ml?

Solve the margin of error for n

$$e = z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}} \quad \Longleftrightarrow \quad \sqrt{n} = \frac{z_{1-\alpha/2}\,\sigma}{e} \quad \Longrightarrow \quad n \;\geq\; \left(\frac{z_{1-\alpha/2}\,\sigma}{e}\right)^2 \quad \text{(round } up\text{)}$$

- n grows with the **square** of $z\sigma/e$: halving the margin of error needs four times as many observations.
- Higher confidence (larger z) or a larger spread σ also cost sample size.
- σ unknown? Plan with an estimate from a pilot study or from past data.

Key takeaway

Decide on the precision you need *before* collecting data – the formula tells you what it costs.

Exercise – how many bottles? (planning the sample size)

Problem statement

The dairy (milk, $\sigma = 3\,\text{ml}$ known) wants to estimate the mean fill volume μ *more precisely*. Quality control asks for a 95 % CI whose margin of error (half width) is at most $e = 1\,\text{ml}$.

a) What is the margin of error with the current sample ($n = 9$)?

b) At least how many bottles are needed for $e \leq 1\,\text{ml}$?

c) How does n change if the confidence level is to be 99 %?

d) How does n change (at 95 %) if the margin is halved to $e \leq 0.5\,\text{ml}$?

Hint: $z_{0.975} = 1.96$, $z_{0.995} = 2.576$; always round n *up*.

Solution (1/2) – planning the sample size

Rule: solve the margin of error $e = z_{1-\alpha/2}\,\sigma/\sqrt{n}$ for n:

$$n \;\geq\; \left(\frac{z_{1-\alpha/2}\,\sigma}{e}\right)^2 \qquad \text{(round up to the next integer).}$$

a) $e = 1.96 \cdot \dfrac{3}{\sqrt{9}} = 1.96 \cdot 1 = \mathbf{1.96\ ml}$ (the CI $[\,999.04\,;\,1002.96\,]$ from before)

b) $n \geq \left(\dfrac{1.96 \cdot 3}{1}\right)^2 = 5.88^2 = 34.57 \implies \mathbf{n = 35}$

c) $n \geq \left(\dfrac{2.576 \cdot 3}{1}\right)^2 = 7.728^2 = 59.72 \implies \mathbf{n = 60}$

In words: b) For ± 1 ml instead of ± 1.96 ml the dairy needs almost four times as many bottles (35 instead of 9). c) More confidence costs sample size (60 instead of 35).

Solution (2/2) – planning the sample size

$$\text{d)} \quad n \geq \left(\frac{1.96 \cdot 3}{0.5}\right)^2 = 11.76^2 = 138.30 \implies \mathbf{n = 139}$$

In words: Half the margin $\Rightarrow$ four times the sample ($139 \approx 4 \cdot 35$): n grows with the **square** of $z\sigma/e$.

The same with Python:

```
import math
from scipy import stats
def n_required(conf, sigma, e):
    z = stats.norm.ppf(1 - (1 - conf) / 2)
    return math.ceil((z * sigma / e) ** 2)
print(n_required(0.95, 3, 1), n_required(0.99, 3, 1), n_required(0.95, 3, 0.5))
```

Output

```
35 60 139
```

▶ **Try it:** notebook, section 2.3 (Colab)

More realistic: the variance is unknown

Problem

In practice σ^2 is almost never known. We must *also estimate* it from the sample.

Estimator of the variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

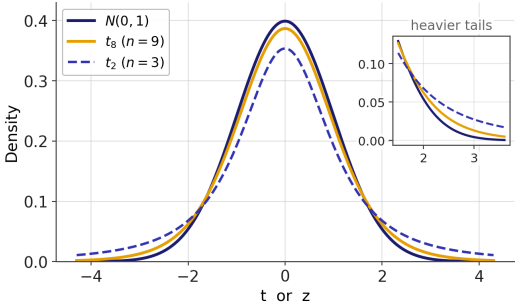
- We replace σ by s in the test statistic.
- This costs *accuracy* – and changes the distribution (see next slide).

Why the t-distribution? – intuitively and graphically

Standardised statistic (normal data)

$$T_n = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

- s is **itself a random variable**.
- T_n is a ratio of *two* random quantities $\Rightarrow$ **heavier tails**.
- For large n, s barely varies, so $t_{n-1} \to N(0,1)$.



Key takeaway

Unknown σ $\Rightarrow$ use t_{n-1}: larger quantiles, so the CI is wider on average.

Milk example with unknown variance

Confidence interval

$$\mathrm{CI} = \left[\bar{X} - t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}},\ \bar{X} + t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}}\right]$$

Data: $\bar{x} = 1001$, $n = 9$, $\alpha = 0.05$, $s = 3.7749$, $t_{0.975;\,8} = 2.3060$.

$$e = 2.3060 \cdot \frac{3.7749}{\sqrt{9}} = 2.9017$$

$$\mathrm{CI}_{0.95} = [\,1001 \pm 2.9017\,] = [\,998.10\,;\,1003.90\,]$$

Comparison

Wider than [999.04 ; 1002.96] (known variance) – the price for estimating σ.

Summary: known vs. unknown variance

	known σ	**unknown σ**
Spread used	σ	s (estimated)
Distribution	$N(0,1)$	t_{n-1}
Quantile (95 %, $n = 9$)	1.96	2.31
CI	narrower	wider (on average)

- For large n ($n \gtrsim 30$) the difference is small ($t_{n-1} \approx N(0,1)$).
- For small samples it is clearly noticeable.

Confidence intervals with Python

```
import numpy as np
from scipy import stats
milk = np.array([1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009])
xbar, n, s = milk.mean(), len(milk), milk.std(ddof=1)     # 1001.0, 9, 3.7749
stats.norm.ppf(0.975), stats.t.ppf(0.975, df=n - 1)      # quantiles: 1.9600, 2.3060
for conf in (0.90, 0.95, 0.99):                           # sigma = 3 known -> N(0,1)
    lo, hi = stats.norm.interval(conf, loc=xbar, scale=3 / np.sqrt(n))
    print(f"z, {conf:.0%}: [{lo:.2f}, {hi:.2f}]")
lo, hi = stats.t.interval(0.95, df=n - 1, loc=xbar, scale=s / np.sqrt(n))
print(f"t, 95%: [{lo:.2f}, {hi:.2f}]")                    # sigma estimated -> t_(n-1)
```

Output

```
z, 90%: [999.36, 1002.64]
z, 95%: [999.04, 1002.96]
z, 99%: [998.42, 1003.58]
t, 95%: [998.10, 1003.90]
```

`interval(conf, ...)` $= \bar{x} \pm$ quantile $\cdot$ SE: `loc` $= \bar{x}$, `scale` = standard error ($\sigma/\sqrt{n}$ or $s/\sqrt{n}$), for t also `df` $= n - 1$. Same numbers as on the milk slides.

▶ **Try it:** notebook, section 2.4 (Colab)

Exercise – filling cream (unknown variance)

Problem statement

The same 16 cream cups as in the known-variance exercise (label 250 g, in g):

$$251, 248, 248, 249, 250, 251, 247, 246, 252, 251, 248, 247, 246, 249, 248, 252$$

This time we do *not* rely on the stated machine variance but estimate σ from the sample. Confidence level $1 - \alpha = 0.95$.

Required: the sample standard deviation s and the confidence interval for μ with *unknown* variance. Does the verdict on 250 g change compared with $[\,247.96\,;\,249.92\,]$ (known variance)?

Hint: $\bar{x} = 248.9375$, $\sum_i (x_i - \bar{x})^2 = 60.9375$, $t_{0.975;\,15} = 2.1314$.

Solution – filling cream (unknown variance)

Rule: as for known variance, but $z \to t_{1-\alpha/2,\,n-1}$ and $\sigma \to s$.
Intermediate values: $s^2 = 60.9375/15 = 4.0625 \Rightarrow s = 2.0156$, $t_{0.975;\,15} = 2.1314$.

$$e = 2.1314 \cdot \frac{2.0156}{\sqrt{16}} = 1.0740$$
$$\mathrm{CI}_{0.95} = [\,248.9375 \pm 1.0740\,] = [\,\mathbf{247.86}\,;\,\mathbf{250.01}\,]$$

Comparison

Now $\mu_0 = 250$ lies (just) *inside* the CI – the verdict turns out *differently* than with known variance. The wider t-interval is more cautious.

Summary and outlook

- Confidence intervals quantify the **uncertainty** of a point estimate.
- Known variance: quantiles of $N(0,1)$; unknown: of t_{n-1}.
- The required precision fixes the **sample size**: $n \geq (z_{1-\alpha/2}\,\sigma/e)^2$.
- **Bridge to the test:** if the CI contains the hypothesised value μ_0, there is no evidence against $H_0 : \mu = \mu_0$ – otherwise there is.

Next section

We formalise this procedure as a **hypothesis test**.

Overview

1 Foundations: a Probability Refresher

2 Confidence Intervals

3 Hypothesis Tests
- Idea and hypotheses
- From CI to test
- Known variance
- Unknown variance
- Tests with Python
- Exercises

4 The p-Value

5 Types of Error

6 Summary

What is a hypothesis test? – intuition

The basic idea, as in court

An assumption (the *null hypothesis* H_0) is held true until the data **speak strongly enough against it**.

- H_0 = "innocent" (status quo).
- We reject H_0 only on *convincing* evidence.
- Little evidence $\Rightarrow$ acquittal for lack of evidence – not "proven innocent".

Milk

$H_0 : \mu = 1000$ – "the machine works correctly". Only a clear deviation from 1000 makes us doubt this.

The two hypotheses

Structure

$$H_0 : \mu = \mu_0 \qquad \text{vs.} \qquad H_1 : \mu \neq \mu_0$$

- We **always** argue under "H_0 is true".
- Why unequal? $H_0 : \mu = \mu_0$ fixes a *single* value, so the test statistic has *one* definite distribution we can compute with; H_1 is a whole *range*, with no single distribution.
- Two outcomes: *keep* H_0 or *reject* it.

Key takeaway

What you want to **establish** statistically goes into H_1.

Which hypothesis goes where?

Rule of thumb

The statement you want to *find evidence for* – the "more important" claim – is formulated as H_1, because only the Type I error (wrongly rejecting H_0) can be controlled via α.

You want evidence that ...	H_0	H_1
the supplier's fill weight *deviates* from the spec	$\mu = \mu_0$	$\mu \neq \mu_0$
the mean processing time *changed* after the update	$\mu = \mu_0$	$\mu \neq \mu_0$
the mean daily revenue *differs* from the plan figure	$\mu = \mu_0$	$\mu \neq \mu_0$

Bridge: confidence interval and test are the same

Testing with the CI

If μ_0 lies *inside* the confidence interval, $H_0 : \mu = \mu_0$ is **not** rejected – if it lies outside, it is.

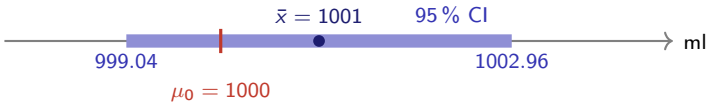
Equivalence – same algebra, two readings

"μ_0 inside the CI" means $|\bar{x} - \mu_0| \leq z_{1-\alpha/2}\,\sigma/\sqrt{n}$, which rearranges to $|Z_n| \leq z_{1-\alpha/2}$ – exactly the non-rejection condition. So a test (level α) and a CI (level $1-\alpha$) **always** agree.

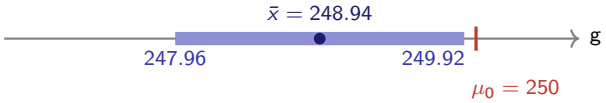
In words: The CI shows *all* μ_0 that you would not reject.

The duality on the number line

Milk: $\mu_0 = 1000$ lies *inside* the CI $\Rightarrow$ **keep** H_0



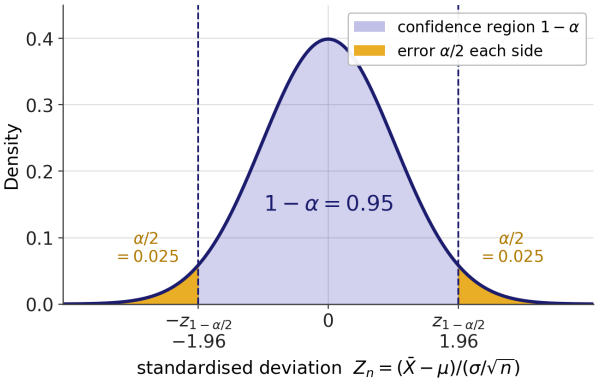
Cream: $\mu_0 = 250$ lies *outside* the CI $\Rightarrow$ **reject** H_0



Key takeaway

μ_0 inside the CI $\Leftrightarrow$ keep H_0; outside $\Leftrightarrow$ reject. (Both at known σ, $\alpha = 0.05$.)

One picture, two readings



- **Distribution reading:** the central blue region holds 95 % of the Z_n distribution (in 95 % of samples Z_n lands here).
- **Test reading:** blue = **acceptance region** (= non-rejection region), orange = **rejection region** (α).
- Test statistic in the orange tail $\Rightarrow$ reject H_0.

"Acceptance region" is conventional for the *non-rejection* region: landing there means we *fail to reject* H_0, not that we accept it as true.

Exercise – hypotheses and deciding with the CI

Problem statement

A bank plans its staff with a mean processing time of $\mu_0 = 30\,\text{min}$ per loan application. Longer times (bottlenecks) and shorter times (overcapacity) matter equally. From experience $\sigma = 6\,\text{min}$ is known. A sample of $n = 36$ applications gives $\bar{x} = 32.8\,\text{min}$; $\alpha = 0.05$.

a) State H_0 and H_1.

b) Compute the 95 % CI for μ.

c) Decide about H_0 with the CI.

d) Which of the planning values $\mu_0 = 31$, 33 and 35 min would a two-sided test at level 5 % reject?

Hint: $z_{0.975} = 1.96$.

Solution – hypotheses and deciding with the CI

Rule: μ_0 inside the $(1-\alpha)$-CI $\Leftrightarrow |Z_n| \le z_{1-\alpha/2} \Leftrightarrow$ keep H_0.

a) $H_0 : \mu = 30$ vs. $H_1 : \mu \neq 30$ – both directions count; the deviation to be detected goes into H_1.

b) $\mathrm{SE} = 6/\sqrt{36} = 6/6 = 1$, $e = 1.96 \cdot 1 = 1.96$,
$\mathrm{CI}_{0.95} = [\,32.8 \pm 1.96\,] = [\,\mathbf{30.84}\,;\,\mathbf{34.76}\,]$ min.

c) $30 \notin \mathrm{CI} \Rightarrow H_0$ is **rejected**: the mean processing time deviates from the planning value.
(Cross-check: $|Z_n| = |32.8 - 30|/1 = 2.8 > 1.96$.)

d) One look at the CI suffices: 31 and 33 lie inside $[\,30.84\,;\,34.76\,] \Rightarrow$ **keep**; 35 lies outside $\Rightarrow$ **reject**.

Key takeaway

A CI answers the two-sided test for *every* hypothesised value μ_0 at once.

The critical value – the bar you must clear

Plain idea

The **critical value** is a fixed threshold on the test statistic. It splits all possible outcomes into two zones:

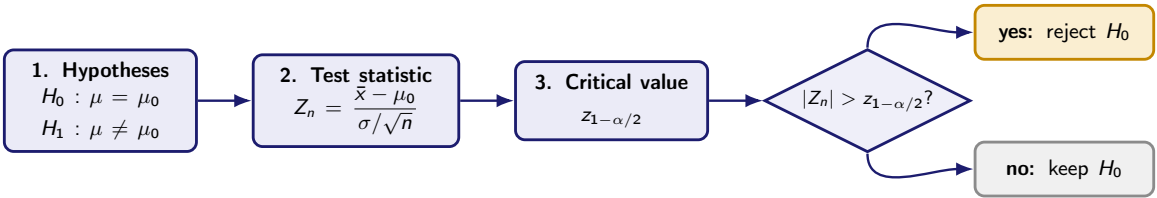
Keep H_0

Statistic *inside* the threshold – the deviation is small enough to pass as ordinary chance.

Reject H_0

Statistic *beyond* the threshold – too extreme to be chance at level α.

The test recipe (known variance)



- **Key move:** plug the hypothesised μ_0 in for the unknown μ (the same Z_n as for the CI, only with μ_0 instead of μ) – *if* H_0 holds then $Z_n \sim N(0,1)$, so a $|Z_n|$ that would be very unusual for $N(0,1)$ is evidence against H_0.
- $|Z_n| \leq z_{1-\alpha/2}$: acceptance (non-rejection) region; beyond it: rejection region (area α under H_0). Fix α *in advance*.

Milk test with known variance

Data: $\bar{x} = 1001$, $\sigma = 3$, $n = 9$, $\mu_0 = 1000$, $\alpha = 0.05$, $z_{0.975} = 1.96$.

$$Z_n = \frac{1001 - 1000}{3/\sqrt{9}} = \frac{1}{1} = 1.000$$

$$|Z_n| = 1.000 \; < \; 1.96 = z_{0.975}$$

Decision

$H_0 : \mu = 1000$ is **not** rejected – no evidence against a correctly working machine.

In words: Same result as the CI: $\mu_0 = 1000 \in [999.04\,;\,1002.96]$.

Test with unknown variance

Only two changes to the recipe

$$T_n = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0, \qquad \text{rejection: } |T_n| > t_{1-\alpha/2,\, n-1}.$$

- $\sigma \to s$ (estimated) and $z \to t_{n-1}$.
- **Assumption:** data (approximately) normally distributed *or* $n \geq 30$ (central limit theorem).

Milk test with unknown variance

Data: $\bar{x} = 1001$, $s = 3.7749$, $n = 9$, $\mu_0 = 1000$, $\alpha = 0.05$, $t_{0.975;\,8} = 2.3060$.

$$T_n = \frac{1001 - 1000}{3.7749/\sqrt{9}} = \frac{1}{1.2583} = 0.7947$$

$$|T_n| = 0.7947 \;<\; 2.3060 = t_{0.975;\,8}$$

Decision

$H_0 : \mu = 1000$ is **not** rejected. (More cautious t-quantile 2.31 instead of 1.96 – the result is nevertheless the same here.)

Before you test – the assumptions checklist

1. **Random sample, independent observations** – the X_i are i.i.d. draws from the population. (No clustering, no repeated measurements of the same unit.)
2. **Distribution:** data approximately normal *or* $n \gtrsim 30$ – then the CLT (Foundations) makes $\bar{X}$ approximately normal.
3. **Spread:** σ known $\rightarrow$ z-test; σ estimated by $s \rightarrow$ t-test with df $= n - 1$.
4. **Question fixed in advance:** μ_0, the direction (one-/two-sided) and α are chosen *before* seeing the results.

Key takeaway

If 1.–2. fail, the stated α is not honest: the 5 % false-alarm guarantee only holds under the model assumptions.

Hypothesis test with Python – `ttest_1samp`

```python
import numpy as np
from scipy import stats
milk = np.array([1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009])
res = stats.ttest_1samp(milk, popmean=1000)     # two-sided by default
ci = res.confidence_interval(confidence_level=0.95)
print(f"t = {res.statistic:.4f}, df = {res.df}, p = {res.pvalue:.4f}")
print(f"95% CI: [{ci.low:.2f}, {ci.high:.2f}]")
```

Output (one-sample t-test)

```
t = 0.7947, df = 8, p = 0.4497
95% CI: [998.10, 1003.90]
```

By default `stats.ttest_1samp` performs a **two-sided** t-test with **unknown** variance; the result holds t, df and p-value, `.confidence_interval()` adds the CI.

▶ **Try it:** notebook, section 3.5 (Colab)

Exercise – cream test

Problem statement

Cream, label 250 g, known variance $\sigma^2 = 4$. Sample $n = 16$, $\bar{x} = 248.9375$. Significance level $\alpha = 0.05$, $\mu_0 = 250$.

Carry out the test – (a) known variance, (b) unknown variance ($s = 2.0156$, $t_{0.975;\,15} = 2.1314$).

Solution (1/2) – cream, known variance

Rule: reject if $|Z_n| > z_{1-\alpha/2}$.

$$Z_n = \frac{248.9375 - 250}{2/\sqrt{16}} = \frac{-1.0625}{0.5} = -2.125$$

$$|Z_n| = 2.125 \; > \; 1.96 = z_{0.975}$$

Decision

$H_0 : \mu = 250$ is **rejected**. There is evidence that the machine does not fill correctly.
(Consistent with the CI: $250 \notin [247.96\,;\,249.92]$.)

Solution (2/2) – cream, unknown variance

$$T_n = \frac{248.9375 - 250}{2.0156/\sqrt{16}} = \frac{-1.0625}{0.5039} = -2.1086$$

$$|T_n| = 2.1086 \; < \; 2.1314 = t_{0.975;\,15}$$

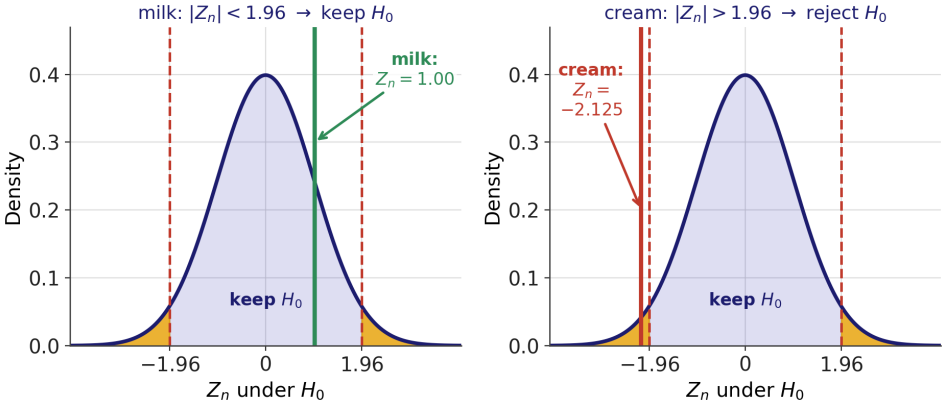
Decision

$H_0 : \mu = 250$ is **not** rejected (just barely!).

Key takeaway

Same data set, a different assumption about σ – a different decision. The t-distribution is more conservative (heavier tails).

Milk vs. cream – the picture behind both decisions



Two-sided test, known σ, $\alpha = 0.05$: the milk statistic ($Z_n = 1.00$) stays *inside* the bars ± 1.96, the cream statistic ($Z_n = -2.125$) lands *beyond* them. The decision is nothing more than: **where does my statistic fall?**

Exercise – reaction time (large sample)

Problem statement

A training programme is meant to change the baseline reaction time $\mu_0 = 250\,\text{ms}$. A sample of $n = 50$ gives $\bar{x} = 242$, $s = 30$. Test *two-sided* at $\alpha = 0.05$.

Hint: $t_{0.975;\,49} = 2.0096$ (already close to 1.96).

Solution – reaction time

$$T_n = \frac{242 - 250}{30/\sqrt{50}} = \frac{-8}{4.2426} = -1.8856, \qquad |T_n| = 1.8856 < 2.0096$$

Decision & outlook

Keep H_0 (two-sided): $p = 2\,F_{t,49}(-1.8856) = \mathbf{0.0653} > 0.05$. With $n = 50 \geq 30$ the CLT justifies the t-test. *Outlook:* had the question – *before* collecting the data – declared only a *shorter* reaction time relevant, a left-sided test would have been appropriate, and it would decide differently here (see the one-sided exercises in the appendix). The direction must never be chosen *afterwards* because $\bar{x} < \mu_0$.

Exercise – delivery time (complete t-test by hand)

Problem statement

A logistics provider contractually guarantees an online shop a **mean delivery time of $\mu_0 = 48$ hours**. The shop's controller wants to check whether the actual mean delivery time *deviates* from this – both directions matter (too slow: customer complaints; too fast: the shop could book a cheaper, slower tariff). For $n = 8$ randomly selected shipments she records the delivery time (in h):

$$51,\ 39,\ 49,\ 41,\ 47,\ 43,\ 45,\ 45$$

Delivery times are approximately normal, σ unknown, $\alpha = 0.05$.

a) Compute $\bar{x}$ and s.
b) State H_0/H_1, compute T_n and decide with the critical value.
c) Compute the 95 % CI for μ and show that it agrees with the test decision.

Hint: $t_{0.975;\,7} = 2.3646$.

Solution (1/4) – delivery time: key figures

Rule: $\bar{x} = \frac{1}{n}\sum x_i, \quad s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2.$

x_i	51	39	49	41	47	43	45	45	$\sum = 360$
$x_i - \bar{x}$	6	-6	4	-4	2	-2	0	0	$\sum = 0$
$(x_i - \bar{x})^2$	36	36	16	16	4	4	0	0	$\sum = 112$

$$\bar{x} = \frac{360}{8} = \mathbf{45\ h}, \qquad s^2 = \frac{112}{8-1} = \frac{112}{7} = 16 \implies s = \mathbf{4\ h}$$

standard error: $\dfrac{s}{\sqrt{n}} = \dfrac{4}{\sqrt{8}} = \dfrac{4}{2.8284} = \mathbf{1.4142\ h}$

In words: The shipments took 45 h on average; single values scatter with 4 h, the sample mean only with 1.41 h.

Solution (2/4) – delivery time: test with the critical value

Hypotheses (two-sided; the direction follows from the question, *not* from the data):

$$H_0 : \mu = 48 \qquad \text{vs.} \qquad H_1 : \mu \neq 48$$

Rule: σ unknown $\Rightarrow$ t-test; reject if $|T_n| > t_{1-\alpha/2;\, n-1} = t_{0.975;\, 7} = 2.3646$.

$$T_n = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{45 - 48}{1.4142} = \frac{-3}{1.4142} = \mathbf{-2.1213}$$

$$|T_n| = 2.1213 \;<\; 2.3646$$

Decision

$H_0 : \mu = 48$ is **not rejected** (narrowly). The observed deviation of 3 h amounts to just over 2 standard errors – with only $n = 8$ shipments this is still compatible with chance; a breach of contract is *not* statistically established.

Solution (3/4) – delivery time: CI and duality

Rule: $\mathrm{CI}_{0.95} = \left[\bar{x} \pm t_{0.975;\,7}\, s/\sqrt{n}\right]$.

$$e = 2.3646 \cdot 1.4142 = 3.344$$
$$\mathrm{CI}_{0.95} = [\,45 \pm 3.344\,] = [\,\mathbf{41.66}\,;\,\mathbf{48.34}\,]\ \mathrm{h}$$

Duality CI $\leftrightarrow$ test

$\mu_0 = 48 \in [\,41.66\,;\,48.34\,] \Leftrightarrow |T_n| \le t_{0.975;\,7} \Leftrightarrow$ keep H_0 – the same algebra, two readings. That 48 lies *just* inside the upper bound mirrors the narrow $|T_n| = 2.12$ vs. 2.36.

In words: Mean delivery times between 41.7 h and 48.3 h are compatible with the data – the contractual 48 h are (just) among them.

Solution (4/4) – delivery time: with Python

```
import numpy as np
from scipy import stats
delivery = np.array([51, 39, 49, 41, 47, 43, 45, 45])     # hours
print(delivery.mean(), delivery.std(ddof=1), round(stats.t.ppf(0.975, df=7), 4))
res = stats.ttest_1samp(delivery, popmean=48)             # two-sided
ci = res.confidence_interval(confidence_level=0.95)
print(f"t = {res.statistic:.4f}, df = {res.df}, p = {res.pvalue:.4f}")
print(f"95% CI: [{ci.low:.2f}, {ci.high:.2f}]")
```

Output ($\bar{x}$, s, critical value; test; CI)

```
45.0 4.0 2.3646
t = -2.1213, df = 7, p = 0.0716
95% CI: [41.66, 48.34]
```

t, df and CI agree with the hand calculation. The reported $p = 0.0716 > 0.05$ leads to the same decision – what it means exactly is the topic of the next section.

▶ **Try it:** notebook, section 3.6 (Colab)

Recap – and one limitation of the recipe

The recipe in one line

Standardise the deviation, compare with the critical value, decide.

Example	Statistic	Bar	Decision	How close?
milk, σ known	$\lvert Z_n \rvert = 1.000$	1.960	keep H_0	clear call
cream, σ known	$\lvert Z_n \rvert = 2.125$	1.960	reject H_0	close call
cream, σ unknown	$\lvert T_n \rvert = 2.109$	2.131	keep H_0	razor-thin

Next section

All three verdicts are a bare *yes/no* – the last column is invisible in them. The p-**value** puts exactly this – the *strength* of the evidence – on a continuous 0–1 scale.

Overview

1 Foundations: a Probability Refresher

2 Confidence Intervals

3 Hypothesis Tests

4 The *p*-Value
- Concept
- Calculation
- Exercises

5 Types of Error

6 Summary

The p-value – intuition

A single question

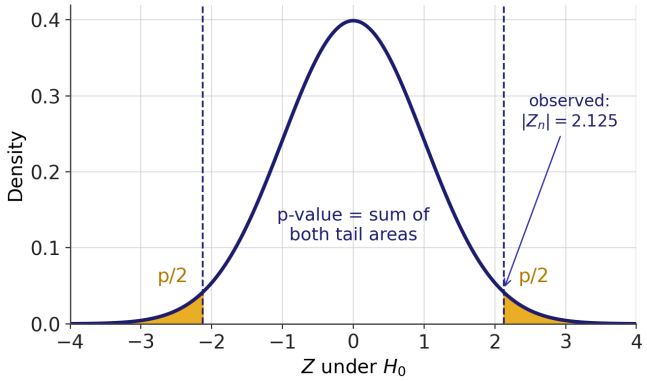
"How **surprising** are my data if H_0 were true?"

- **Small p-value** (≈ 0): such an extreme result would be very rare under $H_0 \Rightarrow$ doubt about H_0.
- **Large p-value** (≈ 1): the data are unsurprising under $H_0 \Rightarrow$ no reason to doubt it (but *not* positive proof of H_0).

Key takeaway

The p-value measures *how surprising* the data are if H_0 holds – the smaller, the stronger the evidence **against** H_0.

The *p*-value – graphically



The (two-sided) *p*-value is the **total area** in both tails that is *at least as extreme* as the observed value.

The p-value – definition

Definition

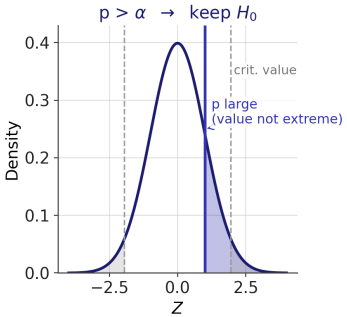
The p-value is the probability, under the validity of H_0, of observing a test statistic that is **at least as extreme** as the one actually observed.

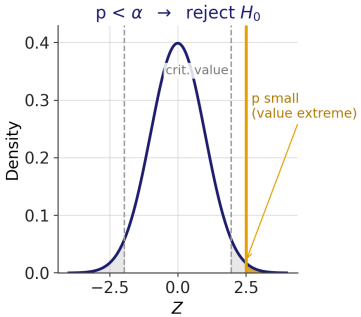
- **"At least as extreme" is set by H_1:** two-sided ($\mu \neq \mu_0$) = both tails $|Z| \geq |Z_n|$; one-sided = only the tail in the H_1 direction. Same statistic, different p.
- $0 \leq p \leq 1$; software (e.g. `scipy.stats`) always reports it.
- **Advantage:** more informative than a bare yes/no – it quantifies the *strength* of the evidence.

Decision rule

One rule for all tests

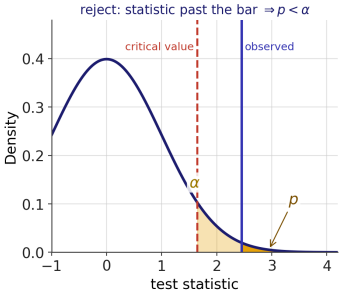
$$p < \alpha \implies \text{reject } H_0 \qquad p \geq \alpha \implies \text{keep } H_0$$

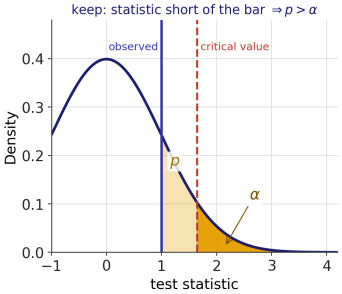




- Left: p large, value in the acceptance region.
- Right: p small, value in the rejection region.
- α is the "benchmark".

Two rules, one decision: critical value $\leftrightarrow$ p-value





- α = tail area beyond the **critical value** (fixed in advance).
- p = tail area beyond your **observed** statistic.
- Statistic past the bar $\Leftrightarrow$ its tail is smaller $\Leftrightarrow$ $p < \alpha$.
- Picture: *right-sided* test ($z_{0.95} = 1.645$); two-sided: both tails, $z_{0.975} = 1.96$.

Key takeaway

The two decision rules are **identical**: "test statistic beyond the critical value" and "$p < \alpha$" always agree – the critical value is simply the statistic whose tail area equals α.

Both rules, one verdict – worked example

Cream, two-sided, known σ: $Z_n = -2.125$, $\alpha = 0.05$.

Critical-value rule

$$|Z_n| = 2.125 > 1.96 = z_{0.975}$$

$$\Rightarrow \textbf{reject } H_0.$$

p-value rule

$$p = 2\,\Phi(-2.125) = 0.034$$

$$0.034 < 0.05 \Rightarrow \textbf{reject } H_0.$$

Key takeaway

Same data, same α, **same decision** – because $|Z_n| > z_{0.975}$ and $p < \alpha$ are just two ways of saying the observed value sits past the bar.

Exercise – p-value and the choice of α

Problem statement

A two-sided test with known variance yields $Z_n = 1.85$. Compute the p-value and decide at $\alpha = 0.05$ and at $\alpha = 0.10$. Check both decisions with the critical-value rule.

Hint: $\Phi(1.85) = 0.9678$ (table: row 1.8, column .05); two-sided as in the worked example: $p = 2\,\Phi(-|Z_n|)$; $z_{0.975} = 1.96$, $z_{0.95} = 1.645$.

Solution – *p*-value and α

Rule: two-sided $p = 2\,\Phi(-|Z_n|) = 2\left(1 - \Phi(|Z_n|)\right)$ (symmetry); reject H_0 if $p < \alpha$.

$$p = 2\left(1 - \Phi(1.85)\right) = 2 \cdot 0.0322 = \mathbf{0.0644} \qquad \text{(exact, unrounded } \Phi\text{: 0.0643)}$$

- At $\alpha = 0.05$: $0.0644 > 0.05 \Rightarrow$ **keep** H_0. Check: $|Z_n| = 1.85 < 1.96 = z_{0.975}$.
- At $\alpha = 0.10$: $0.0644 < 0.10 \Rightarrow$ **reject** H_0. Check: $|Z_n| = 1.85 > 1.645 = z_{0.95}$ (two-sided at 10 % uses $z_{1-0.10/2} = z_{0.95}$).

Key takeaway

The same data can be "significant" or not – α must be fixed *in advance*, never chosen to fit the result.

Computing the p-value (two-sided)

Known variance

$$p = 2 \cdot \Phi\big(-|Z_n|\big) = P\big(|Z| > |Z_n|\big), \quad Z \sim N(0,1)$$

Unknown variance

$$p = 2 \cdot F_{t,n-1}\big(-|T_n|\big)$$

In words: Why double? With $H_1 : \mu \neq \mu_0$ a deviation in *either* direction counts, so "at least as extreme" means *both* tails $|Z| > |Z_n|$; by symmetry each has area $\Phi(-|Z_n|)$ – hence the factor 2. (One-sided: a single tail, no doubling.)

p-value – milk (two variants)

Known variance, $Z_n = 1.000$

$$p = 2 \cdot \Phi(-1.00) = 2 \cdot 0.15866 = 0.3173$$

Unknown variance, $T_n = 0.7947$

$$p = 2 \cdot F_{t,8}(-0.7947) = 2 \cdot 0.22485 = 0.4497$$

Decision

Both p-values $> 0.05 \Rightarrow H_0 : \mu = 1000$ not rejected. The value 0.4497 matches the `ttest_1samp` output.

Exact vs. table: the values above use the unrounded Φ and $F_{t,8}$ (software). From the printed Φ table, $\Phi(1.00) = 0.8413$ gives $p = 2 \cdot 0.1587 = 0.3174$. The t-table can only bracket: $0.7947 < 1.397 = t_{0.90;\,8} \Rightarrow p > 2 \cdot 0.10 = 0.20$.

p-value – cream (two variants)

Known variance, $Z_n = -2.125$

$$p = 2 \cdot \Phi(-2.125)$$
$$= 2 \cdot 0.0168 = \mathbf{0.0336}$$

$0.0336 < 0.05 \Rightarrow$ reject.

Unknown variance, $T_n = -2.1086$

$$p = 2 \cdot F_{t,15}(-2.1086)$$
$$= 2 \cdot 0.0261 = \mathbf{0.0522}$$

$0.0522 > 0.05 \Rightarrow$ keep.

Key takeaway

The same data set can be just *significant* or just *not significant* depending on the assumption about σ.

Exact vs. table: exact values (software). With the printed tables: $\Phi(2.12) = 0.9830$, $\Phi(2.13) = 0.9834 \Rightarrow 0.0332 < p < 0.0340$; t-table, df $= 15$: $1.753 < 2.1086 < 2.131 \Rightarrow 0.05 < p < 0.10$. Both brackets give the same decisions.

p-values with Python

```
import numpy as np
from scipy import stats
milk = np.array([1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009])
cream = np.array([251, 248, 248, 249, 250, 251, 247, 246,
                  252, 251, 248, 247, 246, 249, 248, 252])
for x, mu0, sigma in [(milk, 1000, 3), (cream, 250, 2)]:
    n = len(x)
    z = (x.mean() - mu0) / (sigma / np.sqrt(n))          # sigma known
    t = (x.mean() - mu0) / (x.std(ddof=1) / np.sqrt(n))  # sigma estimated
    p_z = 2 * stats.norm.sf(abs(z))                      # 2 * (1 - Phi(|z|))
    p_t = 2 * stats.t.sf(abs(t), df=n - 1)               # 2 * (1 - F_t,n-1(|t|))
    print(f"z = {z:.4f}, p = {p_z:.4f} | t = {t:.4f}, p = {p_t:.4f}")
```

Output (milk, cream):

```
z = 1.0000, p = 0.3173 | t = 0.7947, p = 0.4497
z = -2.1250, p = 0.0336 | t = -2.1086, p = 0.0522
```

`sf` $= 1 - F$ (survival function). The t-row equals `ttest_1samp(x, mu0)`.

▶ **Try it:** notebook, section 4.2 (Colab)

p-value – typical misconceptions

What the *p*-value is *not*

- **Not** the probability that H_0 is true.
- A large *p*-value does **not prove** H_0 – it only shows *no contradiction*.

Correct – mind the direction

The *p*-value is the probability *of the data (or more extreme)* **assuming** H_0 (chance) holds: $P(\text{data} \mid H_0)$. It *conditions on* chance; it does **not** give the probability that chance was the *cause*, $P(H_0 \mid \text{data})$.

Significant $\neq$ important – the large-n trap

The p-value measures evidence, not size

With enough data, *any* tiny deviation from μ_0 becomes statistically significant – even one that is practically irrelevant.

Milk machine, observed $\bar{x} = 1000.2$ ml ($\mu_0 = 1000$, $\sigma = 3$)

	Z_n	p (two-sided)	verdict at $\alpha = 0.05$
$n = 9$	0.20	0.84	not significant
$n = 10\,000$	6.67	< 0.0001	**highly** significant

Key takeaway

The same irrelevant 0.2 ml deviation – invisible at $n = 9$, "highly significant" at $n = 10\,000$. So report the **effect size and the CI**, not just p.

Exercise – interpret the *p*-value (concept)

Which statements are correct? ($p = 0.03$, $\alpha = 0.05$)

1. "There is a 3 % probability that H_0 is true."
2. "If H_0 were true, data this extreme occur in 3 % of samples."
3. "We reject H_0 at $\alpha = 0.05$."
4. "The effect is large and important."

Solution – interpret the p-value

Answers

1. **Wrong**: p is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$.
2. **Correct**: that is exactly the definition.
3. **Correct**: $0.03 < 0.05$.
4. **Wrong**: significance $\neq$ effect size; a tiny effect can be significant with large n.

Key takeaway

A small p-value signals *evidence against* H_0 – not the probability of H_0, and not the size or importance of the effect.

Exercise – bracketing the p-value with the t-table

Problem statement

Delivery-time exercise (section "Hypothesis tests"): two-sided t-test with $T_n = -2.1213$ and df $= 7$. In the exam there is no software – only the row df $= 7$ of the t-table:

upper-tail probability $P(T > t)$	0.10	0.05	0.025	0.01	0.005
t (df $= 7$)	1.415	1.895	2.365	2.998	3.499

a) Bracket the two-sided p-value from below and above.
b) Decide at $\alpha = 0.05$ and at $\alpha = 0.10$.
c) Which Python command gives the exact value?

Solution – bracketing the *p*-value with the *t*-table

Rule: two-sided $p = 2 \cdot P(T > |T_n|)$; the table gives $P(T > t)$ only at a few points – so *bracket*.

a) $1.895 < |T_n| = 2.121 < 2.365 \implies 0.025 < P(T > 2.121) < 0.05 \implies \mathbf{0.05 < p < 0.10}$

b) $\alpha = 0.05$: $p > 0.05 \Rightarrow$ **keep** H_0 (as $|T_n| < 2.365$). $\alpha = 0.10$: $p < 0.10 \Rightarrow$ **reject** H_0 (as $|T_n| > 1.895$).

c) exactly, with Python

```
from scipy import stats
print(round(2 * stats.t.cdf(-2.1213, df=7), 4))          # exact two-sided p-value: 0.0716
```

Key takeaway

The table yields only an *interval* for p – enough for the decision "$p < \alpha$?"; critical value and *p*-value always agree.

▶ **Try it:** notebook, section 4.3 (Colab)

Exercise – coffee roastery (exam-style, from A to Z)

Problem statement

A coffee roastery fills tins with a target net weight of $\mu_0 = 1000\,\text{g}$. From long-term process monitoring the standard deviation is *known*: $\sigma = 40\,\text{g}$. After a refit, a sample of $n = 64$ tins gives $\bar{x} = 1012\,\text{g}$. Test two-sided at $\alpha = 0.05$.

a) State the hypotheses.
b) Compute the standard error $\sigma/\sqrt{n}$ and the test statistic Z_n.
c) Give the critical value and make the test decision.
d) Compute the p-value with the Φ-table. Which Python command gives it?
e) Compute the 95 % CI for μ. What does it imply for $H_0 : \mu = 1000$?
f) A second check sample gives $n = 25$, $\bar{x} = 1012\,\text{g}$ and an *estimated* $s = 40\,\text{g}$. Compute the 95 % CI ($t_{0.975;\,24} = 2.0639$). Would H_0 be rejected now?

Solution (1/3) – coffee roastery: test with the critical value

a) Deviations in *both* directions matter (too little: consumer protection; too much: cost):

$$H_0 : \mu = 1000 \qquad \text{vs.} \qquad H_1 : \mu \neq 1000$$

b) σ known $\Rightarrow$ z-test (Gauss test):

$$\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{64}} = \frac{40}{8} = \mathbf{5\ g}, \qquad Z_n = \frac{\bar{x} - \mu_0}{\text{SE}} = \frac{1012 - 1000}{5} = \mathbf{2.40}$$

c) Two-sided: $\alpha/2 = 0.025$ in each tail, critical value $z_{0.975} = 1.96$ (table: $\Phi(1.96) = 0.9750$).

Decision

$|Z_n| = 2.40 > 1.96 \Rightarrow H_0$ is **rejected**: the mean fill weight deviates significantly from 1000 g – upwards. Readjusting the machine saves coffee.

Solution (2/3) – coffee roastery: p-value and CI

d) Table: $\Phi(2.40) = 0.9918$, so

$$p = 2\left(1 - \Phi(2.40)\right) = 2 \cdot 0.0082 = \mathbf{0.0164} < 0.05 \;\Rightarrow\; \text{reject } H_0.$$

If the machine were set correctly, a deviation at least this large would occur in only 1.64 % of all samples.

```
from scipy import stats
z = (1012 - 1000) / (40 / 64 ** 0.5)
print(z, round(2 * stats.norm.sf(abs(z)), 4))          # z and two-sided p-value
```

```
2.4 0.0164
```

e) $\quad 1012 \pm 1.96 \cdot 5 = 1012 \pm 9.8 \;\Rightarrow\; \text{CI}_{0.95} = [\,\mathbf{1002.2}\,;\,\mathbf{1021.8}\,]$ g

$1000 \notin \text{CI} \Rightarrow H_0$ is rejected – consistent with $|Z_n| > 1.96$ and $p < 0.05$ (duality).

▶ **Try it:** notebook, section 4.3 (Colab)

Solution (3/3) – coffee roastery: estimated σ

f) Now σ is *estimated* $\Rightarrow$ t-quantile with df $= 24$:

$$\frac{s}{\sqrt{n}} = \frac{40}{5} = 8, \qquad 1012 \pm 2.0639 \cdot 8 = 1012 \pm 16.51 \;\Rightarrow\; [\,\mathbf{995.49}\,;\,\mathbf{1028.51}\,]\ \text{g}$$

	$\sigma = 40$ known, $n = 64$	$s = 40$ estimated, $n = 25$
standard error	$40/8 = 5$	$40/5 = 8$
quantile	$z_{0.975} = 1.96$	$t_{0.975;\,24} = 2.0639$
95 % CI	$[\,1002.2\,;\,1021.8\,]$	$[\,995.49\,;\,1028.51\,]$
contains $\mu_0 = 1000$?	no $\rightarrow$ reject	yes $\rightarrow$ keep

Key takeaway

The second interval is wider for **two** reasons: the smaller sample *and* the larger t-quantile for the estimated spread. Now 1000 lies inside – H_0 would *not* be rejected.

Overview

1 Foundations: a Probability Refresher

2 Confidence Intervals

3 Hypothesis Tests

4 The p-Value

5 Types of Error
- The two types of error
- α and β

6 Summary

Two ways to be wrong

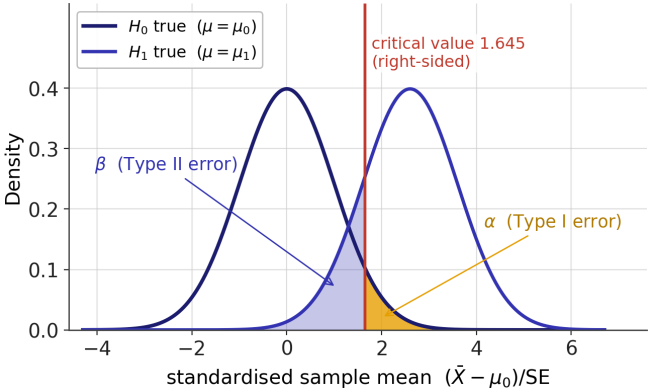
Intuition

Like a smoke detector, we can be wrong in two ways: **false alarm** or **missed fire**.

Decision	H_0 **true**	H_0 **false**
reject H_0	Type I error (α)	correct (power $1-\beta$)
keep H_0	correct ($1-\alpha$)	Type II error (β)

Type I error = false alarm (reject although H_0 is true). **Type II error** = miss (keep although H_1 is true).

α and β in a picture



- The two curves show the *standardised* $\bar{X}$ in each "world": left centred at μ_0 (value 0, if H_0 holds), right at the true μ_1 (here 2.6 standard errors above μ_0, if H_1); the red line is the critical value of a right-sided test (1.645).
- α: H_0 true, but beyond the critical value.
- β: H_1 true, but on this side.
- The distributions overlap – both errors cannot be 0 at once.

Type I error (α) – controllable

Definition

$$P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$$

- **Directly controllable** through the choice of α (typically 0.05).
- In 5 % of cases a *true* H_0 is wrongly rejected.

Milk

$H_0 : \mu = 1000$ is true, yet the test rejects (false alarm) – probability $= \alpha$.

Type II error (β) – only indirectly influenced

Definition (at a specific alternative μ_1)

$$P(\text{keep } H_0 \mid \mu = \mu_1) = \beta$$

- No *direct* lever; β depends on:
  - α, sample size n, spread σ^2,
  - true deviation $|\mu - \mu_0|$ (effect size).
- At a fixed μ_1: smaller $\alpha \Rightarrow$ higher bar $\Rightarrow$ more misses (β *rises*); larger $n \Rightarrow$ smaller SE $\Rightarrow$ β *falls*.

Key takeaway

Power $= 1 - \beta$: the ability to actually reject a *false* H_0.

Only one error is controllable

Question: Which error can we only make when we *reject* H_0?

Answer

The Type I error – and that is exactly the one we control via α.

Consequence for the choice of hypotheses

This is why the claim you want to *establish* goes into H_1 (rule of thumb from above): wrongly declaring it "established" – rejecting H_0 although H_0 holds – is the costly mistake, and exactly this one is the controllable Type I error.

Overview

1 Foundations: a Probability Refresher
2 Confidence Intervals
3 Hypothesis Tests
4 The p-Value
5 Types of Error
6 Summary

Summary 1 – foundations

Normal distribution

$X \sim N(\mu, \sigma^2) \;\Rightarrow\; Z = \dfrac{X-\mu}{\sigma} \sim N(0,1)$

$P(X \le x) = \Phi\left(\frac{x-\mu}{\sigma}\right), \quad \Phi(-z) = 1 - \Phi(z)$

quantile: $x_p = \mu + \sigma\, z_p, \quad z_{0.975} = 1.96$

Sampling distribution of $\bar{X}$

$E(\bar{X}) = \mu, \quad \mathrm{Var}(\bar{X}) = \sigma^2/n, \quad \mathrm{SE} = \sigma/\sqrt{n}$

normal data: $\bar{X} \sim N(\mu, \sigma^2/n)$ exactly;
otherwise (CLT, $n \gtrsim 30$): approximately.

Key takeaway

Every interval and test in this part *standardises* $\bar{X}$ with its standard error – and the CLT makes this work even for non-normal data.

Summary 2 – confidence intervals

Known variance σ $(Z_n \sim N(0,1))$

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

Unknown variance $(T_n \sim t_{n-1})$

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}},\ \bar{X} + t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}}\right]$$

Sample size for a margin of error e (known σ): $n \geq \left(z_{1-\alpha/2}\,\sigma/e\right)^2$, rounded up.

Summary 3 – hypothesis test (two-sided)

$H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$

Test statistics

$$Z_n = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1), \qquad T_n = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \qquad \text{(under } H_0\text{)}$$

Decision

Reject if $|Z_n| > z_{1-\alpha/2}$ (σ known) or $|T_n| > t_{1-\alpha/2,\, n-1}$ (σ unknown).

Summary 4 – the p-value

One idea behind every formula

p = probability, *if* H_0 *were true*, of a value **at least as extreme as observed** – in the direction(s) that H_1 allows.

If H_1 is…	**add up the…**	p **(unknown σ)**
$\mu \neq \mu_0$	*both* tails beyond $\lvert T_n \rvert$	$2\,F_{t,n-1}(-\lvert T_n \rvert)$
$\mu < \mu_0$ (left)	*lower* tail below T_n	$F_{t,n-1}(T_n)$
$\mu > \mu_0$ (right)	*upper* tail above T_n	$1 - F_{t,n-1}(T_n)$

Known σ? Just replace $T_n \to Z_n$ and $F_{t,n-1} \to \Phi$. **Decide:** $p < \alpha \Rightarrow$ reject H_0.

Summary 5 – types of error

Types of error

$P(\text{reject } H_0 \mid \mu = \mu_0) = \alpha \qquad P(\text{keep } H_0 \mid \mu = \mu_1) = \beta$

α is *controllable*; β falls with $n \uparrow$ and $\alpha \uparrow$; power $= 1 - \beta$.

One-sided tests and the power of a test: Appendix A.3–A.8.

Summary 6 – important Python commands

Distributions

```
from scipy import stats
stats.norm.cdf(1.96)            # Phi(z)
stats.norm.sf(1.96)             # 1 - Phi(z)
stats.norm.ppf(0.975)           # z_p
stats.norm.cdf(997, 1000, 3)    # P(X<=x)
stats.t.cdf(-2.11, df=15)       # F_t,df(t)
stats.t.ppf(0.975, df=8)        # t_p,df
```

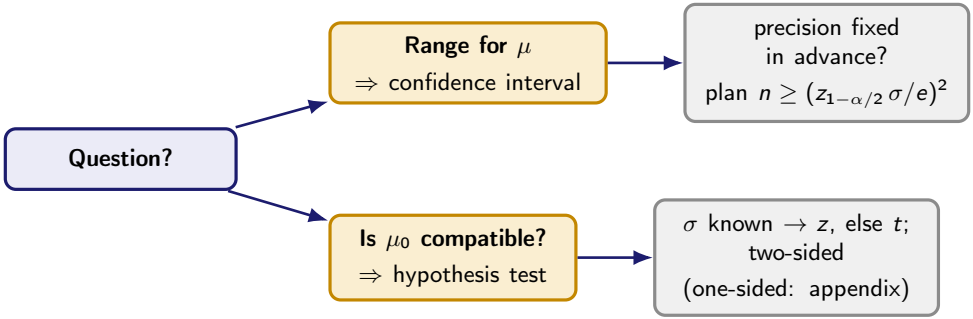
Tests, intervals, sample size

```
res = stats.ttest_1samp(x, popmean=1000)
res = stats.ttest_1samp(x, 1000,
                        alternative="less")
res.statistic, res.df, res.pvalue
res.confidence_interval(0.95)    # CI
2 * stats.norm.sf(abs(z))        # z-test p
math.ceil((1.96 * sigma / e)**2)  # n
```

`alternative` is `"two-sided"` (default), `"less"` or `"greater"`; `scale` in `stats.norm` is the *standard deviation*; `confidence_interval` returns the CI that matches the chosen direction.

▶ **Try it:** notebook, Key Python commands (Colab)

Which procedure? – decision tree



The same logic – *standardise*, then ask whether the deviation is extreme *in the H_1 direction* – underlies essentially every classical test: z, t, proportion, regression coefficient, ANOVA F.

Key takeaway

First clarify the *question*, then choose the tool.

Five classic pitfalls – final checklist

Forbidden sentence ×	**Correct version** ✓
"p is the probability that H_0 is true."	$p = P(\text{data at least this extreme} \mid H_0)$.
"μ lies in *this* CI with 95 % probability."	The *procedure* covers μ in 95 % of samples.
"$p > \alpha$, so H_0 is proven/accepted."	No contradiction found – H_0 is *kept*, not proven.
"The data point left, so I test left-sided."	Direction and α are fixed *before* the data.
"Significant, hence large and important."	Significance $\neq$ size: judge magnitude by the CI.

Outlook: Part 4 – Regression

What we have so far

Confidence intervals and tests for **one** population mean μ: estimate, standard error, critical value – and the p-value.

Next part

We study the **relationship between variables**. The regression slope $\hat{\beta}_1$ is again an estimator with a standard error, so the *same* recipe gives its t-test, its confidence interval and its p-value:

$$T = \frac{\text{estimate} - \text{hypothesised value}}{\text{standard error}}.$$

Key takeaway

Everything in this part carries over one-to-one to the coefficients of a regression model.

Overview

7 Appendix: Optional and Advanced Material
- A.1 Properties of estimators
- A.2 Law of large numbers
- A.3 One-sided tests: motivation
- A.4 Left- and right-sided tests
- A.5 One-sided p-values
- A.6 One-sided confidence intervals
- A.7 Exercises on one-sided tests
- A.8 Power of a test

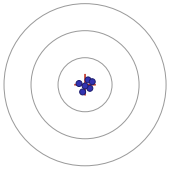
Appendix – what is in here, and why

Optional material

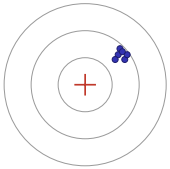
The main part of this deck is self-contained. The appendix adds theory behind it and two extensions of the test: one-sided tests and power.

Topic	**Content**
A.1 Estimators	What makes a good estimator: unbiasedness, efficiency, consistency; why s^2 divides by $n-1$; exercise
A.2 Law of large numbers	Why $\bar{x}$ settles down at μ – and why relative frequencies justify the frequentist view
A.3–A.7 One-sided tests	Direction, critical values, one-sided p-values and CIs; exercises
A.8 Power	β and power $1-\beta$, levers for more power; exercises

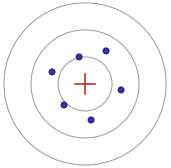
What makes a good estimator? – the target picture



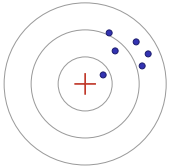
unbiased & efficient



biased & efficient



unbiased & inefficient



biased & inefficient

Each dot = the estimate from *one* sample; red cross = the true value θ. Unbiased = centred on the cross; efficient = tightly clustered.

Properties of estimators – the formal criteria

An **estimator** $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$ is a random variable; its value for the observed sample is the *estimate*.

Property	**Formal definition**
Unbiasedness	$E(\hat{\theta}) = \theta$
Bias	$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$
Asymptotic unbiasedness	$\lim_{n\to\infty} E(\hat{\theta}_n) = \theta$
Efficiency	$\text{Var}(\hat{\theta}) < \text{Var}(\hat{\theta}^*)$ (both unbiased)
Consistency	$\hat{\theta}_n \xrightarrow{P} \theta$, i.e. $\lim_{n\to\infty} P\big(\lvert\hat{\theta}_n - \theta\rvert > c\big) = 0$ for every $c > 0$
sufficient for it	$\lim_{n\to\infty} E(\hat{\theta}_n) = \theta$ and $\lim_{n\to\infty} \text{Var}(\hat{\theta}_n) = 0$

In words: **unbiased** = right on average; **efficient** = small spread; **consistent** = arbitrarily precise as n grows. The sample mean $\bar{X}$ is unbiased and consistent, and the most efficient of all unbiased weighted averages $\sum a_i X_i$.

Why divide by $n-1$ in s^2?

Sample variance – unbiased

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad E(s^2) = \sigma^2.$$

- We measure the spread around the *estimated* mean $\bar{X}$, which sits in the middle of the data – so the squared deviations come out **systematically too small**.
- Dividing by n would give $E\left(\frac{1}{n}\sum(X_i - \bar{X})^2\right) = \frac{n-1}{n}\sigma^2$ – biased downwards; dividing by $n-1$ corrects exactly this.
- Estimating μ "uses up" one degree of freedom – the same $n-1$ as in t_{n-1}.

Simulation: 100 000 samples of size $n = 4$ from $N(50, 5^2)$, i.e. $\sigma^2 = 25$

Average of s^2 with $n-1$: **24.97** vs. with n: **18.73** ($\approx \frac{3}{4} \cdot 25 = 18.75$).

Exercise – properties of estimators: egg weight

Problem statement

A farmer draws $n = 4$ eggs randomly and independently from a flock with $\mu = 50\,\text{g}$ and $\sigma^2 = 25\,\text{g}^2$. Four estimators of μ are proposed:

$$\hat{\mu}_A = \frac{X_1 + X_2 + X_3 + X_4}{4}, \qquad \hat{\mu}_B = \frac{X_1 + X_4}{2},$$

$$\hat{\mu}_C = \frac{X_1 + 2X_2 + X_3 + X_4}{5}, \qquad \hat{\mu}_D = \frac{X_1 + X_2 + X_3}{4}.$$

a) Which estimators are unbiased? Give the expected value and, if any, the bias.

b) Compute the variance of the four estimators.

c) Which estimator should the farmer use? Argue with the criteria.

d) Let n grow ($\hat{\mu}_A = \bar{X}_n$, $\hat{\mu}_B = \frac{X_1 + X_n}{2}$): which of the two is consistent?

Solution (1/2) – egg weight: expected values and variances

Rules: $E\left(\sum a_i X_i\right) = \mu \sum a_i$; for independent X_i: $\mathrm{Var}\left(\sum a_i X_i\right) = \sigma^2 \sum a_i^2$ (here $\mu = 50$, $\sigma^2 = 25$).

	weights a_i	$E(\hat{\mu})$	**Bias**	$\mathrm{Var}(\hat{\mu})$
$\hat{\mu}_A$	$\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$	50	0	$\frac{4}{16} \cdot 25 = \mathbf{6.25}$
$\hat{\mu}_B$	$\frac{1}{2}, 0, 0, \frac{1}{2}$	50	0	$\frac{2}{4} \cdot 25 = \mathbf{12.5}$
$\hat{\mu}_C$	$\frac{1}{5}, \frac{2}{5}, \frac{1}{5}, \frac{1}{5}$	50	0	$\frac{1+4+1+1}{25} \cdot 25 = \mathbf{7}$
$\hat{\mu}_D$	$\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0$	$\frac{3}{4} \cdot 50 = 37.5$	$\mathbf{-12.5}$	$\frac{3}{16} \cdot 25 = \mathbf{4.6875}$

In words: a) A, B, C are unbiased (weights sum to 1); D underestimates the egg weight *systematically* by 12.5 g (weights sum to only $\frac{3}{4}$).

Solution (2/2) – egg weight: choice and consistency

c) Among the *unbiased* estimators, $\hat{\mu}_A = \bar{X}_4$ has the smallest variance – it is **efficient**. $\hat{\mu}_D$ scatters even less but aims systematically at the wrong value: **use $\bar{X}_4$.**

d) $\bar{X}_n$: unbiased and $\text{Var} = \sigma^2/n \to 0 \Rightarrow$ **consistent**. $\frac{X_1+X_n}{2}$: unbiased, but $\text{Var} = \sigma^2/2 = 12.5$ for *every* n (ignores $n-2$ eggs) $\Rightarrow$ **not consistent**.

```
import numpy as np
rng = np.random.default_rng(1)
m = rng.normal(50, 5, size=(10_000, 4))        # 10 000 samples of 4 eggs
A = m.mean(axis=1)
B = (m[:, 0] + m[:, 3]) / 2
C = (m[:, 0] + 2 * m[:, 1] + m[:, 2] + m[:, 3]) / 5
D = (m[:, 0] + m[:, 1] + m[:, 2]) / 4
print(np.round([A.mean(), B.mean(), C.mean(), D.mean()], 2))
print(np.round([A.var(ddof=1), B.var(ddof=1), C.var(ddof=1), D.var(ddof=1)], 2))
```

```
[49.95 49.95 49.95 37.47]
[ 6.14 12.52  6.87  4.52]
```

Simulation and theory agree: means ≈ 50 (except D: 37.5), variances ≈ 6.25, 12.5, 7, 4.69.

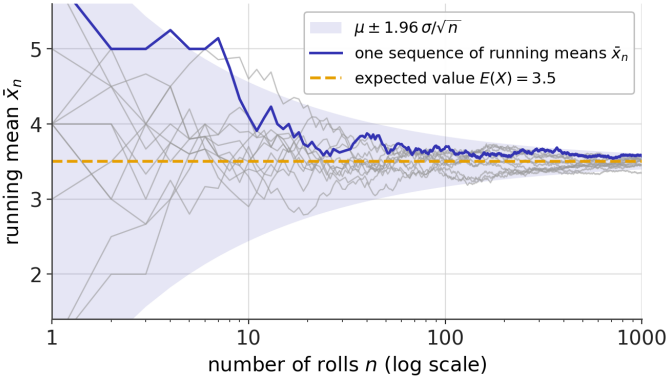
▶ **Try it:** notebook, section A.1 (Colab)

The law of large numbers – intuition

Die example: a fair die has $E(X) = 3.5$. After a few rolls the running mean $\bar{x}_n$ still jumps around; after many rolls it settles down at 3.5.

Key takeaway

The sample mean $\bar{X}_n$ gets ever more **reliably close** to the expected value as n grows.



Twelve sequences of die rolls (grey), one highlighted (blue). The funnel $\mu \pm 1.96\,\sigma/\sqrt{n}$ narrows: ± 1.06 at $n = 10$, ± 0.33 at $n = 100$, ± 0.11 at $n = 1000$.

The law of large numbers – formal statement

Weak law of large numbers

For i.i.d. X_i with $E(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$:

$$\lim_{n\to\infty} P\big(|\bar{X}_n - \mu| > c\big) = 0 \quad \text{for every } c > 0, \qquad \text{i.e.} \quad \bar{X}_n \xrightarrow{P} \mu.$$

In words: Choose any tolerance c, however small: the probability that $\bar{X}_n$ misses μ by more than c goes to zero.
Why? Chebyshev's inequality: $P(|\bar{X}_n - \mu| > c) \le \mathrm{Var}(\bar{X}_n)/c^2 = \sigma^2/(n\,c^2) \to 0$.

Special case: relative frequencies (Bernoulli)

$X_i = 1$ if an event occurs (probability p), else 0: then $\bar{X}_n$ is the event's *relative frequency*, and $\bar{X}_n \xrightarrow{P} p$ – the basis of the **frequentist** view used throughout this part.

LLN: $\bar{X}_n$ approaches μ. CLT: how the remaining fluctuation around μ is distributed.

When only one direction matters

The idea

In a two-sided test *any* departure from μ_0 counts against H_0. Often, though, a deviation is a **problem in only one direction** – the other direction is harmless or even welcome.

Too little in the bottle

For consumer protection only *underfilling* ($\mu < 1000$) is critical – overfilling does not harm the customer.

Unemployment insurance

Only if the rate *exceeds* a threshold do contributions fall short. A lower rate causes no problem.

Key takeaway

The *question* – not the data – decides the direction. The deviation we truly care about defines H_1, and the *whole* α goes on that one side instead of $\alpha/2$ on each.

Business examples – one direction only

You want evidence that ...	H_1	Direction
the new ad campaign *raises* the conversion rate	$\mu > \mu_0$	right-sided
the new process *lowers* the average unit cost	$\mu < \mu_0$	left-sided

In both cases $H_0 : \mu = \mu_0$, as before.

Key takeaway

As in the two-sided case, the claim goes into H_1 – but now its *inequality* also fixes the side on which we reject.

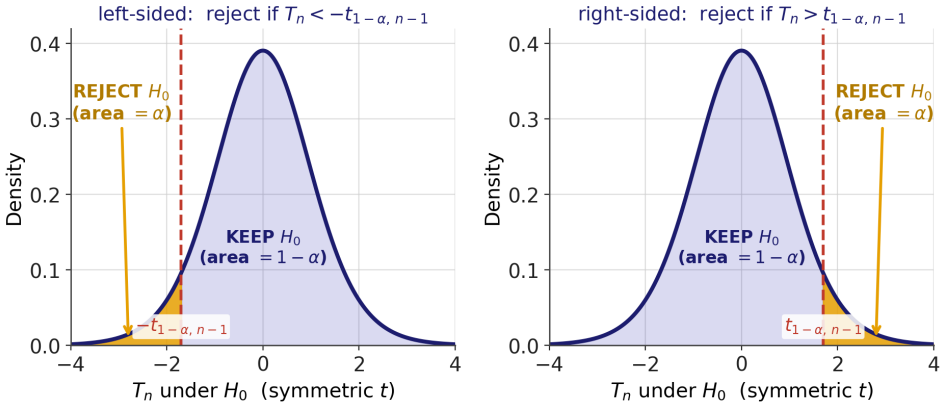
How to find the critical value – 3 steps

1. **Direction** (read from H_1): one-sided or two-sided?
2. **Tail probability:**
   - two-sided $\rightarrow$ *split* α: use the $1 - \alpha/2$ quantile ($\pm$ value),
   - one-sided $\rightarrow$ keep α *whole*: use the $1 - \alpha$ quantile (one value, on the H_1 side).
3. **Which table?** σ known $\rightarrow$ z (normal); σ unknown $\rightarrow$ t with df $= n - 1$.

$\alpha = 0.05$ – the resulting critical values

known σ: two-sided $z_{0.975} = 1.96$, one-sided $z_{0.95} = 1.645$ (Φ table).
unknown σ, $n = 12$: two-sided $t_{0.975;\,11} = 2.20$, one-sided $t_{0.95;\,11} = 1.80$ (`stats.t.ppf`; the t-table follows).

One-sided critical values – left and right

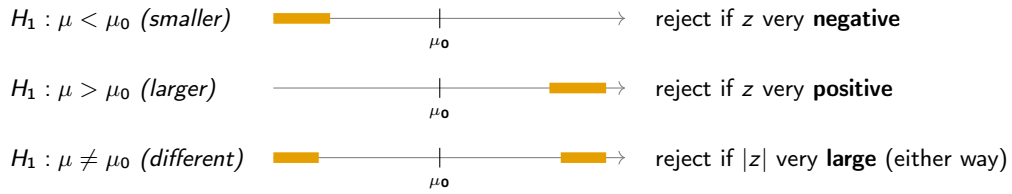


The whole rejection area α sits in *one* tail – left for "too small", right for "too large". The curve is the **symmetric** t-distribution: only the side changes, the shape never does.

The golden rule of one-sided tests

If you remember only one sentence

We reject H_0 when the sample result is **sufficiently extreme in the direction specified by the alternative hypothesis** H_1.



Key takeaway

The inequality in H_1 points straight at the rejection region (orange): "$<$" $\rightarrow$ left tail, "$>$" $\rightarrow$ right tail, "$\neq$" $\rightarrow$ both.

A left-sided test as a lawyer's argument

Your claim ($H_1 : \mu < 10$): "the true mean is *smaller* than 10." The sample mean $\bar{x}$ is your evidence:

$\bar{x} = 8$	$\bar{x} = 9$	$\bar{x} = 10$	$\bar{x} = 11$	$\bar{x} = 12$
strong *for*	some *for*	neutral	against	against

- Below 10 $\Rightarrow$ evidence *for* H_1; if extreme enough, **reject** H_0.
- $\bar{x} = 12$ is far from 10 but in the *wrong* direction – it gives **no** support for "$\mu < 10$".

Key takeaway

The test asks whether the evidence *in the* H_1 *direction* is strong enough to reject H_0 – a big deviation the wrong way counts for nothing.

Critical values of the standard normal $N(0,1)$

Confidence $1-\alpha$	α	**one-sided** $z_{1-\alpha}$	**two-sided** $z_{1-\alpha/2}$
90 %	0.10	1.282	1.645
95 %	0.05	1.645	1.960
98 %	0.02	2.054	2.326
99 %	0.01	2.326	2.576
99.9 %	0.001	3.090	3.291

In words: Pattern: the two-sided value at level α equals the one-sided value at level $\alpha/2$ – e.g. two-sided 95% = 1.96 = one-sided 97.5%.

Critical values of the t-distribution t_{df}

	upper-tail probability (one-sided α | two-sided $\alpha/2$)				
$\text{df} = n - 1$	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
15	1.341	1.753	2.131	2.602	2.947
20	1.325	1.725	2.086	2.528	2.845
30	1.310	1.697	2.042	2.457	2.750
60	1.296	1.671	2.000	2.390	2.660
∞	1.282	1.645	1.960	2.326	2.576

Highlighted: $t_{0.975;\,9} = 2.262$ (two-sided 95%, $n = 10$); the ∞-row is the normal.

How to read the tables – worked lookups

Three questions, every time

(1) one- or two-sided? (2) which α? (3) z or t (df $= n - 1$)?

Normal (σ known):

- two-sided, $\alpha = 0.05 \to$ column $1 - \frac{\alpha}{2} = 0.975$ $\to z = 1.96$.
- one-sided, $\alpha = 0.05 \to 1 - \alpha = 0.95$ $\to z = 1.645$.

t (σ unknown), $n = 10 \Rightarrow$ df$= 9$:

- two-sided, $\alpha = 0.05 \to$ upper-tail 0.025 $\to t = 2.262$.
- one-sided, $\alpha = 0.05 \to$ upper-tail 0.05 $\to t = 1.833$.

Key takeaway

Two-sided *halves* the tail probability ($\alpha \to \alpha/2$); the t-row is df $= n - 1$. As df grows, the t-value slides towards z.

Exercise – test direction and critical value

Problem statement

For each case, state H_1, the test direction and the rejection rule with the critical value from the $N(0,1)$- or t-table of the previous slides (always $H_0 : \mu = \mu_0$).

1. A parcel service wants to show that its mean delivery time is *below* 24 h. σ unknown, $n = 16$, $\alpha = 0.05$.
2. An energy supplier checks whether the mean annual consumption per household has *risen* above last year's 3 500 kWh. σ known, $\alpha = 0.01$.
3. A brewery checks whether the mean fill volume *deviates* from the target of 500 ml. σ unknown, $n = 10$, $\alpha = 0.01$.
4. An online shop wants to show that a new layout *increases* the mean time on site from 4 min so far. σ unknown, $n = 21$, $\alpha = 0.10$.

Solution – test direction and critical value

Rule: (1) direction from the wording $\rightarrow H_1$; (2) one-sided: quantile $1-\alpha$, two-sided: $1-\alpha/2$; (3) σ known $\rightarrow z$, otherwise t with df $= n-1$.

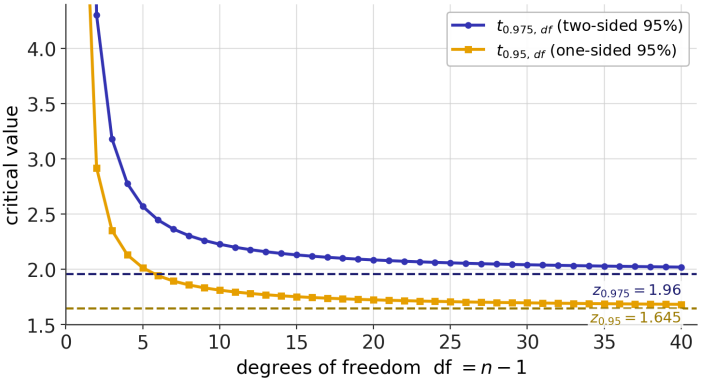
	H_1	Direction	Table value	Reject H_0 if ...
1	$\mu < 24$	left-sided	$t_{0.95;\,15} = 1.753$	$T_n < -\mathbf{1.753}$
2	$\mu > 3\,500$	right-sided	$z_{0.99} = 2.326$	$Z_n > \mathbf{2.326}$
3	$\mu \neq 500$	two-sided	$t_{0.995;\,9} = 3.250$	$\lvert T_n \rvert > \mathbf{3.250}$
4	$\mu > 4$	right-sided	$t_{0.90;\,20} = 1.325$	$T_n > \mathbf{1.325}$

Upper-tail probability in the t-table: one-sided the whole α (cases 1 and 4: columns 0.05 and 0.10), two-sided $\alpha/2 = 0.005$ (case 3). Case 2: one-sided 99 % row of the $N(0,1)$ table.

Key takeaway

First the direction, then α (split or not), then the table (z, or t with df $= n-1$).

Why the t-table ends in the normal



- Small df $\rightarrow$ heavier tails $\rightarrow$ *larger* critical value.
- As df grows, the t-value shrinks towards the z-value.
- By df ≈ 30 the gap is tiny – the ∞ row *is* the normal.

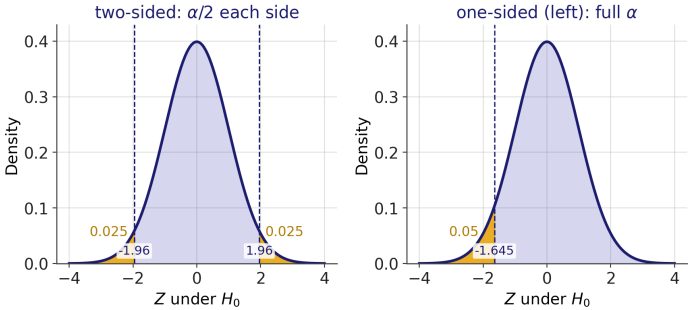
The wording reveals the test direction

Signal words	Hypotheses	Direction
exceed, greater, increased	$H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$	right-sided
fall below, smaller, reduced	$H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$	left-sided

Note

The *question* decides one- or two-sided – not the data. The statement you want to find evidence for goes into H_1.

Why put the whole α on one side?



- Two-sided spends $\alpha/2$ guarding *each* tail – guarding the harmless side is wasted.
- One-sided puts the *whole* α in the relevant tail, so the critical value moves inward ($1.96 \to 1.645$).
- A true deviation in that direction now clears the bar *more easily*.

Key takeaway

Same α, more power – but *only* in the chosen direction. A one-sided test at level α uses the critical value of a two-sided test at level 2α on that side.

The price: you must commit in advance

The direction is part of the hypothesis

Choose one- vs. two-sided **before** looking at the data – based on the research question alone.

- Picking the side *after* seeing which way the data point inflates the true Type I error from α to as much as 2α.
- A one-sided test is **blind** to deviations in the other direction: even a large effect on the "wrong" side never rejects H_0.
- If both directions genuinely matter, use a two-sided test.

In words: One-sided tests buy power with a promise – that the other direction really does not matter.

Left-sided test – consumer protection

Task

A new filling line fills milk (label 1 l). Are the consumer protection requirements met, i.e. is *not too little* being filled? Test with $1 - \alpha = 0.95$.

- $H_0 : \mu = 1000$ vs. $H_1 : \mu < 1000$.
- Only $\mu < 1000$ (too little in the bottle) is critical.

Procedure

Test statistic $T_n = (\bar{X} - \mu_0)/(s/\sqrt{n})$; reject if $T_n < -t_{1-\alpha,\, n-1}$ (known: $Z_n < -z_{1-\alpha}$).
p-value: $p = F_{t,n-1}(T_n)$.

In words: We reject only if $\bar{X}$ falls *clearly below* μ_0 – a value above μ_0 can never be evidence for "too little".

Right-sided test – do not exceed a limit

Task

Advertisement: "Our milk contains **at most** $1\,\text{mg} = 1000\,\mu\text{g}$ of substance K per litre." Is this met? We measure in μg, so $\mu_0 = 1000\,\mu\text{g}$.

- $H_0 : \mu = 1000\,\mu\text{g}$ vs. $H_1 : \mu > 1000\,\mu\text{g}$.
- *Exceeding* the limit is critical.

Procedure

Reject if $T_n > t_{1-\alpha,\,n-1}$ (known: $Z_n > z_{1-\alpha}$).
p-value: $p = 1 - F_{t,n-1}(T_n)$.

In words: Mirror image of the left-sided case: we reject only for values *clearly above* μ_0.

Right-sided test – substance K with numbers

Data (known variance): $\sigma = 90\,\mu\text{g}$, $n = 9$, $\bar{x} = 1060\,\mu\text{g}$, $\mu_0 = 1000$, $\alpha = 0.05$, $z_{0.95} = 1.645$.

$$Z_n = \frac{1060 - 1000}{90/\sqrt{9}} = \frac{60}{30} = 2.00$$

$$Z_n = 2.00 \;>\; 1.645 = z_{0.95}$$

Decision

$H_0 : \mu = 1000$ is **rejected** ($p = 1 - \Phi(2.00) = 0.0228$). Evidence that the limit is *exceeded* – the advertising claim is not statistically tenable.

Left vs. right-sided – at a glance

	left-sided	**right-sided**
Question	"too *small*?"	"too *large*?"
H_0 / H_1	$\mu = \mu_0$ / $\mu < \mu_0$	$\mu = \mu_0$ / $\mu > \mu_0$
Reject if	$T_n < -t_{1-\alpha,n-1}$	$T_n > t_{1-\alpha,n-1}$
Rejection region	far *left* tail	far *right* tail
p-value	$F_{t,n-1}(T_n)$	$1 - F_{t,n-1}(T_n)$

Some textbooks write the null as $\mu \geq \mu_0$ (left-sided) or $\mu \leq \mu_0$ (right-sided) – the test is **identical**, because critical value and p-value are computed at the boundary $\mu = \mu_0$ either way.

Key takeaway

H_1 always points *towards* the rejection region: the inequality in H_1 tells you which tail you reject in.

Exercise – discount campaign (right-sided t-test)

Problem statement

An online shop wants to show that a discount campaign *raises* the mean basket value above its previous level of $\mu_0 = 48$ EUR. A sample of $n = 25$ orders placed during the campaign gives $\bar{x} = 51.20$ EUR and $s = 8$ EUR; $\alpha = 0.05$.

a) State H_0 and H_1. Which test direction?

b) Determine the critical value and the rejection rule.

c) Compute T_n and decide.

d) From which sample mean $\bar{x}$ onwards would the shop have rejected H_0?

Hint: $t_{0.95;\,24} = 1.7109$.

Solution – discount campaign (right-sided t-test)

Rule: right-sided: reject H_0 if $T_n > t_{1-\alpha;\,n-1}$; $T_n = (\bar{x} - \mu_0)/(s/\sqrt{n})$.

a) "raises ... above" $\Rightarrow$ $H_0 : \mu = 48$ vs. $H_1 : \mu > 48$ (**right-sided**).

b) σ unknown $\Rightarrow$ t-test, df $= 25 - 1 = 24$: reject if $T_n > t_{0.95;\,24} = \mathbf{1.7109}$.

c) $\mathrm{SE} = s/\sqrt{n} = 8/\sqrt{25} = 8/5 = 1.6$; $T_n = \dfrac{51.20 - 48}{1.6} = \dfrac{3.2}{1.6} = \mathbf{2.00} > 1.7109$.

d) Solve $T_n > 1.7109$ for $\bar{x}$: $\bar{x}_{\mathrm{crit}} = \mu_0 + t_{0.95;\,24} \cdot \mathrm{SE} = 48 + 1.7109 \cdot 1.6 = \mathbf{50.74}$ EUR.

Decision

H_0 is **rejected**: there is evidence that the campaign raises the mean basket value – any sample mean above 50.74 EUR would have been enough.

One-sided p-value – formulas

Left-sided

$$p = F_{t,n-1}(T_n)$$

known: $p = \Phi(Z_n)$

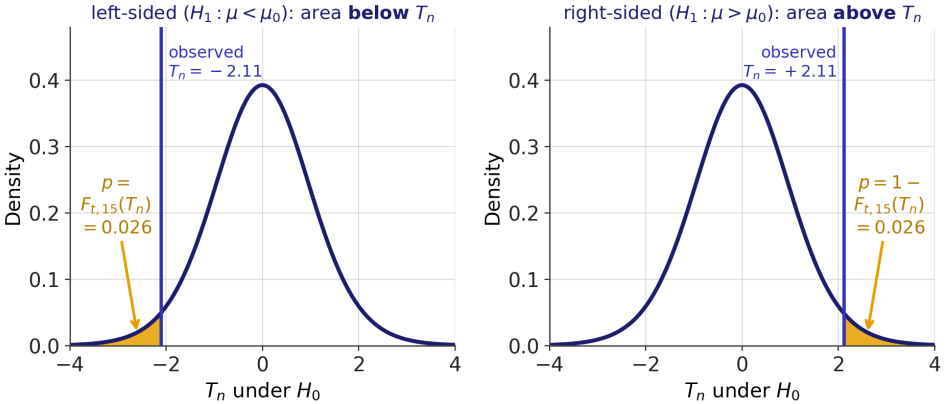
Right-sided

$$p = 1 - F_{t,n-1}(T_n)$$

known: $p = 1 - \Phi(Z_n)$

In words: In a one-sided test we do *not* double – only the one relevant tail area counts.

One-sided p-value – graphically



The p-value is the *single* tail area beyond the observed statistic, **in the direction of** H_1. Left panel: the cream example ($T_n = -2.11$, df $= 15$, $p = 0.026$).

One-sided p-value – mind the sign

Only the H_1-tail counts

The one-sided p-value is the probability, under H_0, of a value *at least as far in the H_1 direction* as the one observed. The opposite tail is irrelevant – a deviation there supports H_0.

Sign check – a big $|z|$ in the wrong direction is *not* significant

Right-sided test with $z = -2.5$: the statistic is on the H_0 side, so $p = 1 - \Phi(-2.5) = 0.994 > 0.5$ – you **never** reject, even though $|z|$ is large. Check the data point in the H_1 direction *before* computing.

In words: Link to two-sided: if the deviation is in the H_1 direction, $p_{\text{one}} = p_{\text{two}}/2$; otherwise $p_{\text{one}} = 1 - p_{\text{two}}/2$.

One-sided p-value – milk ($T_n = 0.7947$, df $= 8$)

Left-sided

$$p = F_{t,8}(0.7947) = 0.7751$$

$> 0.05 \Rightarrow H_0 : \mu = 1000$ kept.

Right-sided

$$p = 1 - 0.7751 = 0.2249$$

$> 0.05 \Rightarrow H_0 : \mu = 1000$ kept.

Interpretation (left)

No evidence that the machine fills *too little* – from a statistical point of view, consumer protection is met.

One-sided p-value – cream (left, unknown var.)

With $T_n = -2.1086$, df $= 15$:

$$p = F_{t,15}(-2.1086) = 0.0261$$

Decision

$p = 0.0261 < 0.05 \Rightarrow H_0 : \mu = 250$ is **rejected** (in favour of $H_1 : \mu < 250$).

Key takeaway

In contrast to the two-sided test ($p = 0.0522$), the left-sided test is **significant** here – because the whole α lies on the critical side.

One-sided test with Python

```
stats.ttest_1samp(milk, 1000, alternative="less")       # H1: mu < mu0, p = 0.7751
stats.ttest_1samp(milk, 1000, alternative="greater")    # H1: mu > mu0, p = 0.2249

cream = np.array([251, 248, 248, 249, 250, 251, 247, 246,
                  252, 251, 248, 247, 246, 249, 248, 252])
res = stats.ttest_1samp(cream, 250, alternative="less")    # left-sided
ci = res.confidence_interval(0.95)                         # one-sided CI
print(f"t = {res.statistic:.4f}, df = {res.df}, p = {res.pvalue:.4f}")
print(f"95% CI: [{ci.low}, {ci.high:.2f}]")
```

Cream, left-sided – output

```
t = -2.1086, df = 15, p = 0.0261
95% CI: [-inf, 249.82]
```

The argument `alternative` controls the direction (`"two-sided"` (default), `"less"`, `"greater"`).

▶ **Try it:** notebook, section A.5 (Colab)

Exercise – one- vs. two-sided on the same data

Problem statement

A coating must *exceed* a target thickness. A test gives $T_n = 1.80$ with df $= 15$. Compare the right-sided and the two-sided decision at $\alpha = 0.05$ – with the p-value and with the critical value.

Hint: $1 - F_{t,15}(1.80) = 0.0460$; t-table, row df $= 15$.

Solution – one- vs. two-sided

Rule: one-sided p = the single tail in the H_1 direction; two-sided p = both tails = $2\times$ the one-sided value (here T_n points in the H_1 direction).

$$\text{right-sided: } p = 1 - F_{t,15}(1.80) = \mathbf{0.0460} < 0.05 \;\Rightarrow\; \text{reject}$$
$$\text{two-sided: } p = 2 \cdot 0.0460 = \mathbf{0.0920} > 0.05 \;\Rightarrow\; \text{keep}$$

Check with the critical values (t-table, row df = 15): right-sided $T_n = 1.80 > 1.753 = t_{0.95;\,15} \Rightarrow$ reject; two-sided $|T_n| = 1.80 < 2.131 = t_{0.975;\,15} \Rightarrow$ keep – the same verdicts.

Key takeaway

Identical data, opposite decisions. The one-sided test is more powerful *in the chosen direction* – exactly why the direction must be fixed before seeing the data.

One-sided confidence intervals

- A one-sided test has a one-sided CI – only *one* bound, the other is $\pm\infty$.
- `scipy` reports them with `low=-inf` or `high=inf`, i.e. $(-\infty;\, x]$ or $[x;\, +\infty)$.

Calculation (known variance)

$$\text{left-sided test } (H_1 : \mu < \mu_0): \quad \left(-\infty,\ \bar{X} + z_{1-\alpha}\frac{\sigma}{\sqrt{n}}\right]$$

$$\text{right-sided test } (H_1 : \mu > \mu_0): \quad \left[\bar{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}},\ +\infty\right)$$

For unknown variance: replace $z_{1-\alpha}$ by $t_{1-\alpha,\, n-1}$.

Note the one-sided quantile $z_{1-\alpha}$ (not $z_{1-\alpha/2}$) – the whole α sits on one side, exactly as in the test.

One-sided CIs – milk and cream (known)

CI for the *left-sided* test: upper bound ($z_{0.95} = 1.645$):

$$\text{Milk: } 1001 + 1.645 \cdot \frac{3}{\sqrt{9}} = 1002.645$$

$$\text{Cream: } 248.9375 + 1.645 \cdot \frac{2}{\sqrt{16}} = 249.760$$

CI for the *right-sided* test: lower bound:

$$\text{Milk: } 1001 - 1.645 \cdot \frac{3}{\sqrt{9}} = 999.355$$

$$\text{Cream: } 248.9375 - 1.645 \cdot \frac{2}{\sqrt{16}} = 248.115$$

In words: If μ_0 lies beyond the single bound, H_0 is rejected.

Exercise – net weight (left-sided, unknown σ)

Problem statement

A package must contain *at least* $\mu_0 = 500\,\text{g}$. A sample of $n = 12$ gives $\bar{x} = 496$, $s = 6$. Is there evidence of underfilling? ($\alpha = 0.05$.)

Hint: $t_{0.95;\,11} = 1.7959$.

Solution – net weight

Underfilling is critical $\Rightarrow H_0 : \mu = 500$ vs. $H_1 : \mu < 500$ (left-sided). Reject if $T_n < -t_{0.95;\,11} = -1.7959$.

$$T_n = \frac{496 - 500}{6/\sqrt{12}} = \frac{-4}{1.7321} = -2.3094 < -1.7959$$

Decision

Reject H_0: evidence of underfilling. $p = F_{t,11}(-2.3094) = \mathbf{0.0207} < 0.05$.

Exercise – muesli filling (exam-style, one-sided t-test)

Problem statement

A plant fills muesli into packs with target weight $\mu_0 = 250\,\text{g}$. Overfilling costs money, so management asks: does the plant fill *more* than 250 g on average? After maintenance, a sample of $n = 25$ packs gives $\bar{x} = 252.6\,\text{g}$ and $s = 5.0\,\text{g}$; $\alpha = 0.05$.

Table values: $t_{0.95;\,24} = 1.7109$, $t_{0.975;\,24} = 2.0639$, $t_{0.99;\,24} = 2.4922$, $t_{0.995;\,24} = 2.7969$; $\Phi(2.60) = 0.9953$.

a) State the hypotheses.
b) Compute the standard error, the test statistic and the critical value, and decide.
c) Which Python command performs this test for a data vector `x`?
d) Bracket the p-value with the t-table, and approximate it with the normal distribution.
e) Compute the (two-sided) 95 % CI for μ.
f) Decide *without new calculation* whether $H_0 : \mu = 253$ is rejected by a two-sided test at $\alpha = 0.05$.

Solution (1/3) – muesli: hypotheses and test

a) The claim to be established (overfilling) goes into $H_1 \Rightarrow$ **right-sided**:

$$H_0 : \mu = 250 \qquad \text{vs.} \qquad H_1 : \mu > 250$$

b) σ unknown $\Rightarrow$ t-test with df $= n - 1 = 24$:

$$\frac{s}{\sqrt{n}} = \frac{5.0}{\sqrt{25}} = \mathbf{1.0\ g}, \qquad T_n = \frac{252.6 - 250}{1.0} = \mathbf{2.60}, \qquad \text{critical value } t_{0.95;\,24} = \mathbf{1.7109}$$

One-sided: the *whole* α sits in the right tail, so the 0.95-quantile (not 0.975).

Decision

$T_n = 2.60 > 1.7109 \Rightarrow H_0$ is **rejected**: the plant fills significantly more than 250 g on average – it should be readjusted.

Solution (2/3) – muesli: Python and the p-value

c) and the exact p-value from the summary figures:

```
from scipy import stats
# raw data x available:  stats.ttest_1samp(x, popmean=250, alternative="greater")
t = (252.6 - 250) / (5.0 / 25 ** 0.5)                    # from the summary figures
print(f"{t:.2f} {stats.t.sf(t, df=24):.4f} {stats.norm.sf(t):.4f}")
```

```
2.60 0.0079 0.0047
```

d) T_n lies between the 0.99- and the 0.995-quantile:

$$t_{0.99;\,24} = 2.4922 \;<\; 2.60 \;<\; 2.7969 = t_{0.995;\,24} \;\Longrightarrow\; \mathbf{0.005 < p < 0.01}$$

One-sided: the upper-tail areas 0.01 and 0.005 *are* the bounds – no doubling (exact: $p = 0.0079$).

Normal approximation: $p \approx 1 - \Phi(2.60) = 1 - 0.9953 = \mathbf{0.0047}$ – smaller than the exact 0.0079, because the t-distribution has heavier tails. Both are far below 0.05: same decision.

▶ **Try it:** notebook, section A.7 (Colab)

Solution (3/3) – muesli: CI and duality

e) $252.6 \pm 2.0639 \cdot 1.0 = 252.6 \pm 2.0639 \;\Rightarrow\; \text{CI}_{0.95} = [\,\mathbf{250.54}\,;\,\mathbf{254.66}\,]$ g

f) The two-sided CI contains exactly the values μ_0 that a two-sided test does *not* reject:

$$250.54 < 253 < 254.66 \;\Rightarrow\; H_0 : \mu = 253 \text{ is } \textbf{not} \text{ rejected.}$$

Every $\mu_0 < 250.54$ or $\mu_0 > 254.66$ would be rejected – e.g. $\mu_0 = 250$.

Careful with the duality

The *two-sided* CI belongs to the *two-sided* test. The partner of the right-sided test in b) is the one-sided CI $[252.6 - 1.7109 \cdot 1.0\,;\,\infty) = [250.89\,;\,\infty)$ – it excludes 250, consistent with the rejection.

Exercise – delivery time revisited (one-sided test vs. two-sided CI)

Problem statement

The logistics provider from the delivery-time exercise now advertises a mean delivery time *below* $\mu_0 = 48$ hours. A new sample of $n = 16$ shipments gives $\bar{x} = 46.4\,\text{h}$ and $s = 3.2\,\text{h}$; $\alpha = 0.05$.

Table values: $t_{0.95;\,15} = 1.7531$, $t_{0.975;\,15} = 2.1314$.

a) State the hypotheses and carry out the test with the critical value.

b) Compute the two-sided 95 % CI. Does it contain 48? Is that a contradiction to a)?

c) Which confidence interval matches the test in a)? Compute it.

d) Check a) and c) with Python.

Solution (1/2) – delivery time revisited: left-sided test

a) The advertised claim goes into $H_1 \Rightarrow$ **left-sided**:

$$H_0 : \mu = 48 \qquad \text{vs.} \qquad H_1 : \mu < 48$$

$$\frac{s}{\sqrt{n}} = \frac{3.2}{\sqrt{16}} = 0.8, \qquad T_n = \frac{46.4 - 48}{0.8} = \mathbf{-2.00}, \qquad \text{critical value: } -t_{0.95;\,15} = \mathbf{-1.7531}$$

Decision

$T_n = -2.00 < -1.7531 \Rightarrow H_0$ is **rejected** ($p = F_{t,15}(-2.00) = 0.0320$): the mean delivery time is significantly below 48 h – the advertisement is statistically supported.

The negative sign belongs to the result: the rejection region lies on the *left*. Comparing absolute values would lose the direction.

Solution (2/2) – delivery time revisited: which CI?

b) $46.4 \pm 2.1314 \cdot 0.8 = 46.4 \pm 1.7051 \Rightarrow \text{CI}_{0.95} = [\,\mathbf{44.69}\,;\,\mathbf{48.11}\,]$ – it *contains* 48! *No* contradiction: the two-sided CI belongs to the *two-sided* test, which would indeed keep H_0 here ($p = 0.064$). The one-sided test puts the whole α on one side and is sharper in that direction.

c) The partner of the left-sided test is the one-sided CI with an upper bound:

$$\left(-\infty\,;\; 46.4 + 1.7531 \cdot 0.8\right] = (-\infty\,;\,\mathbf{47.80}] \quad \Rightarrow \quad 48 \text{ lies outside – consistent with a).}$$

d)

```
from scipy import stats
t = (46.4 - 48) / (3.2 / 16 ** 0.5)
print(f"{t:.2f} {stats.t.cdf(t, df=15):.4f}")            # T_n, left-sided p-value
print(f"{46.4 + stats.t.ppf(0.95, df=15) * 0.8:.2f}")    # upper bound of one-sided CI
```

```
-2.00 0.0320
47.80
```

With raw data: `stats.ttest_1samp(x, 48, alternative="less")` and `.confidence_interval(0.95)`.

▶ **Try it:** notebook, section A.7 (Colab)

One-sided tests at a glance

One-sided tests (unknown variance)

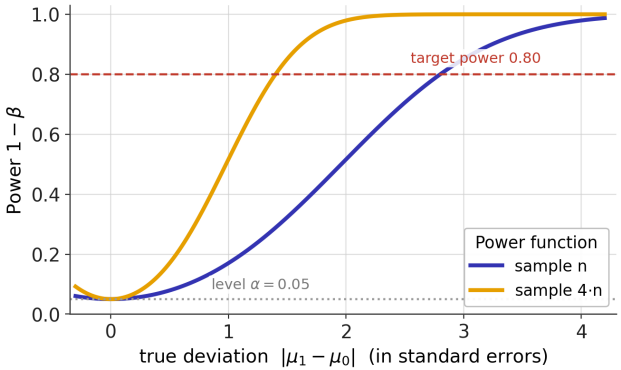
left: $H_1 : \mu < \mu_0$ reject if $T_n < -t_{1-\alpha,\, n-1}$

right: $H_1 : \mu > \mu_0$ reject if $T_n > t_{1-\alpha,\, n-1}$

Key takeaway

Fix the direction *before* looking at the data – the whole α then sits on one side.

Trade-off and power



- $\alpha \downarrow \Rightarrow \beta \uparrow$ (trade-off).
- Larger n raises power at every $\mu \neq \mu_0$ (at μ_0 it stays at α) – more power at the same level.
- Power rises with the true deviation.

Practice

Typically $\alpha = 0.05$ and desired power $1 - \beta \geq 0.80$. A **power analysis** provides the required n.

Computing β concretely – milk

Setup: $H_0 : \mu = 1000$ vs. $H_1 : \mu \neq 1000$, $\sigma = 3$, $n = 9$ (SE $= 1$), $\alpha = 0.05$.

Step 1 – turn "$|Z_n| \leq 1.96$" into bounds on $\bar{X}$:
$\mu_0 \pm 1.96 \cdot \text{SE} = 1000 \pm 1.96 = [998.04\,;\, 1001.96]$.

Step 2 – the truth is $\mu_1 = 1002$; find how often $\bar{X}$ still lands in that region, *standardising with* μ_1:

$$\begin{aligned}\beta &= P\big(998.04 \leq \bar{X} \leq 1001.96 \;\big|\; \mu = 1002\big) \\ &= \Phi\Big(\tfrac{1001.96-1002}{1}\Big) - \Phi\Big(\tfrac{998.04-1002}{1}\Big) = 0.484\end{aligned}$$

Result

Power $= 1 - \beta = 0.516$: a true deviation of 2 ml is detected with $n = 9$ in only just over half the cases.

Four levers for more power

More sample size helps decisively. Same situation with $n = 36$ instead of 9:

$$\text{SE} = \frac{3}{\sqrt{36}} = 0.5 \;\Rightarrow\; \beta = 0.021, \quad \text{power} = 0.979.$$

Levers of power $1 - \beta$

1. **Sample size** $n \uparrow$ (strongest lever).
2. **Effect size** $|\mu_1 - \mu_0|$ (given).
3. **Spread** $\sigma \downarrow$ (e.g. measure more precisely).
4. **Level** $\alpha \uparrow$ (buys more Type I error).

Key takeaway

Quadrupling n from 9 to 36 raises the power from 0.52 to 0.98.

β and power with Python

```
import numpy as np
from scipy import stats
from statsmodels.stats.power import NormalIndPower
for n in (9, 36):                          # milk: mu0 = 1000, true mu1 = 1002, sigma = 3
    se = 3 / np.sqrt(n)
    lo, hi = 1000 - 1.96 * se, 1000 + 1.96 * se          # acceptance region (H0 world)
    beta = stats.norm.cdf(hi, 1002, se) - stats.norm.cdf(lo, 1002, se)  # mu1 world
    print(f"n = {n}: beta = {beta:.3f}, power = {1 - beta:.3f}")
pw = NormalIndPower()                      # power of the z-test; ratio=0: one sample
pw.power(effect_size=2/3, nobs1=9, alpha=0.05, ratio=0)              # 0.516
pw.solve_power(effect_size=2/3, power=0.80, alpha=0.05, ratio=0)     # 17.66 -> n = 18
```

Output

```
n = 9: beta = 0.484, power = 0.516
n = 36: beta = 0.021, power = 0.979
```

Effect size $d = |\mu_1 - \mu_0|/\sigma = 2/3$. `solve_power` does the power analysis: 80 % power needs at least $n = 18$ bottles.

▶ **Try it:** notebook, section A.8 (Colab)

Exercise – compute β and power (two-sided)

Problem statement

Test $H_0 : \mu = 100$ vs. $H_1 : \mu \neq 100$, known $\sigma = 10$, $n = 25$, $\alpha = 0.05$. The true mean is actually $\mu_1 = 105$. Find the Type II error β and the power.

Step 1: the acceptance region for $\bar{X}$ under H_0.

Solution – β and power (two-sided)

SE $= 10/\sqrt{25} = 2$. Acceptance region: $100 \pm 1.96 \cdot 2 = [96.08\,;\,103.92]$.

$$\begin{aligned}\beta &= P(96.08 \le \bar{X} \le 103.92 \mid \mu = 105) \\ &= \Phi\Big(\tfrac{103.92-105}{2}\Big) - \Phi\Big(\tfrac{96.08-105}{2}\Big) = \Phi(-0.54) - \Phi(-4.46) \\ &= 0.2946 - 0.0000 = \mathbf{0.2946}\end{aligned}$$

Power

$1 - \beta = \mathbf{0.7054}$: a true shift of 5 units is detected in about 71 % of samples.

Exercise – power of a one-sided test

Problem statement

Now use a *right-sided* test, $H_1 : \mu > 100$, with $\sigma = 10$, $n = 25$, $\alpha = 0.05$, true $\mu_1 = 105$. Find the power and compare with the two-sided value 0.7054.

Solution – one-sided power

SE $= 2$. Critical mean: $\bar{x}_{\text{crit}} = 100 + 1.645 \cdot 2 = 103.29$.

$$1 - \beta = P(\bar{X} > 103.29 \mid \mu = 105) = 1 - \Phi\left(\tfrac{103.29 - 105}{2}\right) = 1 - \Phi(-0.855) = \mathbf{0.8037}$$

Comparison

Power 0.804 vs. 0.705 two-sided – the one-sided test is *more powerful* for the same n and α, because the whole α sits on the relevant side.